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Analysis and Forecasting of Connectedness Between Financial Markets and the Cryptocurrency Market: A Hybrid Approach Based on QVAR and Machine Learning

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Dedication

I dedicate this work to myself, for my perseverance in the face of adversity and my ability to rise again after every fall. This is not merely an academic achievement, but the result of a journey marked by sacrifices and hard work. I have managed to turn my dreams into reality despite the obstacles along the way. Today, I am proud of the journey I have completed with courage and determination.

To my dearest mother

No matter what I do or say, I will never be able to thank you enough. Your affection protects me, your kindness guides me, and your presence by my side has always given me the strength to overcome difficulties. Thank you for your endless love.

To my dearest father

You have always been by my side, supporting and encouraging me. May this work express my gratitude and affection for you. Your trust and advice have guided me throughout my journey. This achievement is yours as well. Thank you for your quiet, unwavering love.

.

To all my friends

In memory of the wonderful moments we have shared, I dedicate this work to you, wishing you a bright future filled with promise and success.

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List of Abbreviations

AI : Artificial Intelligence

APT : Arbitrage Pricing Theory

BTC : Bitcoin

CAPM : Capital Asset Pricing Model

CMA : Conservative Minus Aggressive

COVID-19 : Coronavirus Disease 2019

DCC : Dynamic Conditional Correlation

DXY: U.S. Dollar Index

ETF : Exchange Traded Fund

ETH : Ethereum

FEVD : Forecast Error Variance Decomposition

GARCH : Generalized Autoregressive Conditional Heteroskedasticity

HME : Hypothèse des Marchés Efficients

IF: Isolation Forest

MCoP: Maximum Connectedness Optimized Portfolio

MCP : Maximum Sharpe Portfolio

MEDAF : Modèle d'Évaluation Des Actifs Financiers

ML : Machine Learning

MVP: Minimum Variance Portfolio

QVAR : Quantile Vector Autoregression

RF : Random Forest

RMW : Robust Minus Weak

S&P 500 : Standard & Poor's 500

TCI : Total Connectedness Index

USDC : USD Coin

USDT : Tether USD

VAR : Vector Autoregression

VaR : Value at Risk

WTI : West Texas Intermediate

General introduction

Over the past decades, financial markets have undergone profound transformations driven by financial globalization, technological innovation, and the acceleration of international capital flows. These developments have strengthened interdependencies between different asset classes, making markets increasingly sensitive to economic, financial, geopolitical, and health-related shocks. Recent events, such as the 2008 global financial crisis, the COVID-19 pandemic, the Russia–Ukraine conflict, and the tightening of monetary policies by major central banks, have demonstrated that disruptions affecting one market can rapidly spread to other segments of the global financial system. In this context, the analysis of financial connectedness has become a major concern for researchers, investors, and regulatory authorities, as it helps improve understanding of risk transmission mechanisms and supports the development of more effective portfolio management strategies.

At the same time, cryptocurrencies have gradually emerged as a new asset class. Since the introduction of Bitcoin in 2009, followed by the development of Ethereum and numerous other digital currencies, their market capitalization, liquidity, and institutional adoption have continued to grow. Initially considered independent of traditional financial markets, cryptocurrencies now appear to maintain increasingly close relationships with stock markets, commodities, currencies, and other financial assets. This evolution raises several questions regarding their role in portfolio diversification, their contribution to systemic risk, and the changing nature of their interactions with traditional financial markets.

In addition, cryptocurrency markets are exposed to several specific types of risk. Regulatory risk represents a major concern, as certain jurisdictions may impose restrictions or even prohibitions on the use or exchange of digital assets. Technological risk is another important source of uncertainty, particularly because of potential vulnerabilities in smart contracts and the possibility of so-called “51% attacks.” Liquidity risk is also significant, especially for low-market-capitalization altcoins, where bid–ask spreads can be substantial. Finally, market manipulation remains a major concern, particularly through practices such as wash trading and pump-and-dump schemes.

In this context, Gandal et al. (2018) provided empirical evidence of price manipulation in the Bitcoin market during the 2013–2014 period, particularly through the use of automated trading bots. These findings highlight the need for an in-depth analysis of price formation mechanisms and the microstructure of cryptocurrency markets.

The study of connectedness between traditional financial markets and cryptocurrencies is therefore particularly important. However, most existing studies have focused primarily on the static or dynamic analysis of interdependencies between markets, while relatively few have proposed an integrated framework combining connectedness measurement, artificial intelligence-based forecasting, anomaly detection, and an assessment of the impact of these forecasts on optimal asset allocation. Recent developments in machine learning offer new opportunities to improve forecasting accuracy and better anticipate regime changes in financial markets.

This research proposes an innovative methodological framework based on several complementary approaches. First, dynamic connectedness among eight financial markets is estimated using the QVAR Connectedness approach, which measures shock transmission across different levels of market risk. The resulting connectedness indices are then used in a Random Forest model to identify the main determinants of financial connectedness. An anomaly-detection procedure is subsequently incorporated to improve the quality of the training data and enhance the predictive performance of an improved Random Forest model. Finally, the connectedness forecasts are integrated into a portfolio optimization process to evaluate their contribution to the construction of more efficient and better-diversified portfolios.

Within this framework, the main research problem can be formulated as follows:

To what extent can a hybrid model combining financial connectedness analysis, machine learning techniques, and anomaly detection improve the forecasting of interdependencies between traditional financial markets and cryptocurrencies, as well as the optimization of investment decisions?

This problem leads to the following research question:

Can the integration of a hybrid model based on dynamic connectedness, Random Forest, and Isolation Forest generate more reliable financial connectedness forecasts and improve investment portfolio performance compared with conventional approaches?

To answer this question, the study pursues the following objectives:

- To conduct a static and dynamic analysis of connectedness between selected financial assets and cryptocurrencies, particularly Bitcoin and Ethereum.

- To develop a hybrid model combining QVAR, Random Forest, and Isolation Forest to improve forecasts of the Total Connectedness Index (TCI).
- To use the resulting forecasts to construct and compare several portfolio optimization strategies, evaluating their performance in terms of returns, risk, and diversification.

To achieve these objectives, the dissertation is organized into three complementary chapters.

The first chapter focuses on the theoretical, conceptual, and empirical foundations of the research. It presents the fundamental characteristics of cryptocurrencies and the main financial theories used in this study, particularly Modern Portfolio Theory and the Capital Asset Pricing Model (CAPM).

The second chapter presents the methodology and empirical results. It also includes a literature review of empirical studies examining connectedness between traditional financial markets and cryptocurrencies, the econometric approaches used to measure shock transmission, and recent research applying machine learning techniques to financial forecasting. This review identifies the limitations of existing studies and highlights the contribution of the present research.

The chapter then describes the data used and the different stages of the proposed methodology, including the estimation of connectedness through the QVAR Connectedness approach, the application of Random Forest and Isolation Forest, and the development of the selected hybrid model. It subsequently presents the results, their financial interpretation, and their implications for portfolio optimization, before outlining the main recommendations and future research directions.

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CHAPTER 1 : Theoretical and Conceptual Framework

Introduction

The theoretical framework of this research adopts a multidimensional approach to analyzing the relationships between cryptocurrencies and traditional financial markets. The rise of digital assets has profoundly transformed the global financial landscape by creating new investment opportunities, new forms of speculation, and increasingly significant interactions between different markets. This transformation raises several questions, particularly regarding the degree to which cryptocurrencies are integrated into the traditional financial system, their impact on market stability, and their ability to improve portfolio diversification.

Analyzing this issue requires the use of several complementary theoretical frameworks. First, it is necessary to examine how traditional financial markets operate and identify the specific characteristics of cryptocurrency markets, whose development is driven by major technological innovations, such as blockchain, as well as substantial speculative activity.

Second, classical financial theories provide an essential analytical framework for understanding price formation mechanisms, investment decisions, market efficiency, and the relationship between risk and return. However, these approaches do not always fully explain certain behaviors observed in contemporary financial markets, particularly in cryptocurrency markets, which are characterized by high volatility and increased sensitivity to investor reactions. In this context, insights from behavioral finance are essential for complementing the analysis.

Furthermore, examining the relationships between cryptocurrencies and traditional financial markets naturally leads to the concept of financial connectedness. This concept makes it possible to analyze how shocks, risks, and volatility fluctuations are transmitted from one market to another. Understanding these mechanisms is essential for examining interdependencies between different assets, assessing contagion risks, and evaluating their implications for financial system stability and portfolio management strategies.

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Section 1. Cryptocurrency Markets

1.1. Origins and Evolution of Cryptocurrencies

The Bitcoin network was officially launched on January 3, 2009, with the creation of the first block, known as the Genesis Block. This event marked the beginning of a new stage in the evolution of digital financial systems. Through this innovation, Satoshi Nakamoto introduced a digital currency independent of central authorities, relying on cryptography to ensure transaction security and the transparency of the entire system.

Since then, the cryptocurrency market has experienced substantial growth, both in the number of projects and in overall market capitalization. What initially began as a technological innovation intended for a relatively small community has evolved into a global financial market attracting an increasing number of retail investors, financial institutions, multinational corporations, and investment funds. Today, thousands of digital assets coexist, each designed to meet specific needs or serve particular purposes.

The evolution of this ecosystem is generally described in terms of several generations of cryptocurrencies. The first generation is represented by Bitcoin, which was primarily designed as a decentralized digital currency offering an alternative to traditional monetary systems. Its main objective is to facilitate secure transfers of value between users without requiring an intermediary. However, certain technical limitations, particularly its limited programmability, gradually encouraged the emergence of a new generation of projects.

The second generation is represented by Ethereum, launched in 2015 by Vitalik Buterin. Unlike Bitcoin, Ethereum extends beyond the basic function of a digital currency by introducing smart contracts. These are computer programs capable of automatically executing predefined contractual conditions once the required criteria have been met, without human intervention. This innovation enabled the development of decentralized applications (dApps) and significantly expanded the potential applications of blockchain technology.

Over the years, the cryptocurrency ecosystem has incorporated numerous innovations, resulting in an increasingly diverse environment. Among the most important developments are decentralized finance (DeFi), which provides financial services such as lending, savings, exchange, and insurance without traditional banking intermediaries, and non-fungible tokens (NFTs), which are designed to certify ownership of unique digital assets. Stablecoins also play an important role in this ecosystem. Because their value is generally pegged to relatively stable assets, such as the US dollar, they aim to reduce the high volatility associated with conventional cryptocurrencies.

Today, Satoshi Nakamoto's white paper is recognized as the foundational text of cryptocurrency economics. It paved the way for a profound transformation of financial systems and digital technologies, the effects of which continue to stimulate academic research and regulatory debate. Cryptocurrencies now raise major questions concerning financial stability, regulatory frameworks, monetary sovereignty, and the growing interactions between traditional financial markets and digital asset markets.

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1.2. Technical Architecture: Blockchain Technology

Blockchain is the fundamental technology underlying the vast majority of cryptocurrencies. It is defined as a distributed, transparent, and immutable ledger in which transactions are recorded in the form of successive blocks. Each block is cryptographically linked to the previous block, thereby forming a chain of blocks.

This structure provides the system with several essential properties, including decentralization, which eliminates the need for a central authority; transparency, which allows transactions to be publicly verified; immutability, which ensures that validated transactions cannot be modified or deleted; and security, which is provided by cryptographic mechanisms such as the SHA-256 algorithm used in the Bitcoin network.

Blockchain operates primarily through consensus mechanisms, which are a central component of its architecture. Proof of Work (PoW), used by Bitcoin, requires network nodes known as miners to solve complex mathematical problems in order to validate transactions and create new blocks. In return, they receive a reward in the form of cryptocurrency.

By contrast, Proof of Stake (PoS), adopted by Ethereum following the The Merge upgrade in September 2022, relies on a validation mechanism based on token ownership and the commitment of tokens as collateral through staking. These two approaches differ significantly in terms of energy consumption, security, and the degree of decentralization.

In the academic literature, Yermack (2015) examines the economic nature of Bitcoin through the traditional functions of money: the unit of account, the medium of exchange, and the store of value. His findings show that Bitcoin performs these functions to a limited extent

and resembles a speculative asset more than a genuine currency in the traditional economic sense..

1.3. Structure and Participants in Cryptocurrency Markets

Cryptocurrency markets differ fundamentally from traditional financial markets in several major structural characteristics.

First, they operate continuously, 24 hours a day, seven days a week, and 365 days a year, without interruptions related to public holidays or stock exchange opening hours. This continuous operation increases their responsiveness to information shocks but also makes them more sensitive to episodes of extreme volatility.

Second, these markets are generally decentralized, although a significant proportion of trading volume is concentrated on centralized exchanges (CEXs), such as Binance, Coinbase, and Kraken. Finally, the regulatory framework remains fragmented and is still developing in many jurisdictions. This creates considerable legal uncertainty and differences in the rules applied internationally.

Participants in cryptocurrency markets are diverse and operate at different levels of the ecosystem. They include miners and validators, who are responsible for ensuring the security and integrity of blockchain networks; centralized and decentralized exchanges (DEXs), such as Uniswap; institutional investors, including specialized hedge funds and venture capital firms; and retail investors.

Another category consists of “whales,” meaning individuals or entities holding very large amounts of crypto assets and capable of significantly influencing market prices and liquidity. Finally, stablecoin issuers, such as Tether and Circle, occupy an increasingly central position within the ecosystem.

Stablecoins now play a systemic role in cryptocurrency markets by serving as instruments of liquidity and internal stores of value. Their growing importance makes them a key component of system stability, but also a potential source of systemic risk. A loss of confidence or a breakdown in a stablecoin’s peg could trigger significant contagion effects across the entire market, as illustrated by the collapse of Terra/LUNA in May 2022.

1.4. Distinctive Characteristics and Specific Risks

Cryptocurrencies differ from traditional financial assets in terms of their risk–return profile. Their historical volatility is considerably higher than that observed in stock or bond markets. For example, Bitcoin has experienced daily price movements exceeding 20% during certain periods of severe instability, whereas the annualized volatility of the S&P 500 index generally ranges between 15% and 25% under normal market conditions. This substantial instability complicates the application of traditional risk-management models, particularly Value at Risk (VaR) models based on the assumption that returns follow a normal distribution.

Furthermore, cryptocurrency markets are exposed to several specific types of risk. Regulatory risk is a major factor, as certain jurisdictions may impose restrictions or even prohibit the use or exchange of digital assets.

Technological risk is another important source of uncertainty, particularly because of potential vulnerabilities in smart contracts or the possibility of so-called “51% attacks.” In addition, liquidity risk is especially significant for low-market-capitalization altcoins, where bid–ask spreads can be substantial.

Finally, market manipulation risk remains a major concern, particularly through practices such as wash trading and pump-and-dump schemes..

1.5. Key Cryptocurrency Market Indicators

Several indicators can be used to assess the state and dynamics of cryptocurrency markets. Among them, total market capitalization (Total Market Cap) is the principal overall indicator, as it aggregates the market value of all cryptocurrencies in circulation and provides a general overview of the cryptocurrency ecosystem.

Another widely used indicator is the Bitcoin Dominance Index, which measures Bitcoin’s share of the total cryptocurrency market capitalization. This indicator is often interpreted as a signal of risk appetite within the cryptocurrency ecosystem, since high Bitcoin dominance generally reflects a shift toward the sector’s most established asset, which is perceived as relatively less risky.

Furthermore, the Crypto Fear & Greed Index, developed by Alternative.me, combines several dimensions of market sentiment into a score ranging from 0 to 100. These dimensions include volatility, trading volumes, social media activity, and certain sentiment surveys. The index therefore provides a useful proxy for measuring the overall emotional state of investors in cryptocurrency markets.

These indicators play an important role in financial connectedness analysis because they make it possible not only to observe price co-movements but also to identify sentiment transmission between cryptocurrency markets and traditional financial markets.

In this context, Baker and Wurgler (2006), in a seminal article published in the Journal of Finance, highlighted the decisive role of investor sentiment in explaining stock returns. This conclusion is particularly relevant to cryptocurrency markets, which are highly sensitive to investor behavior and expectations.

Section 2. Classical Finance Theory (Market Efficiency)

2.1. The Efficient Market Hypothesis (EMH)

The Efficient Market Hypothesis (EMH), formulated by Eugene Fama in his seminal 1970 article published in the Journal of Finance, is one of the fundamental pillars of modern financial theory. According to Fama (1970), a market is considered efficient when the prices of financial assets fully and immediately reflect all available information.

This definition has a major implication: if all available information is already incorporated into asset prices, investors cannot systematically earn returns above the market, known as abnormal returns, solely by relying on publicly available information.

Fama distinguishes three forms of market efficiency according to the type of information incorporated into prices:

- **Weak-form efficiency**: Prices incorporate all information contained in historical price and trading-volume data. Under this form of efficiency, technical analysis cannot consistently generate returns above the market.

- **Semi-strong-form efficiency**: Prices reflect all publicly available information, including financial statements, macroeconomic announcements, and monetary policy decisions.

Consequently, neither technical analysis nor fundamental analysis can consistently outperform the market.

- **Strong-form efficiency**: Prices incorporate both public and private or insider information. This implies that no investor, including those with access to privileged information, can consistently earn abnormal returns.

Fama's (1970) article, Efficient Capital Markets: A Review of Empirical Work, remains one of the most influential contributions to modern finance and continues to shape a substantial part of financial economics research.

In cryptocurrency markets, the question of market efficiency remains particularly controversial. Urquhart (2016), in a study published in Economics Letters, tests the weak form of the EMH for Bitcoin and concludes that the market was inefficient over the 2010–2016 period because statistical regularities were present in its returns.

However, subsequent research, notably Nadarajah and Chu (2017), also published in Economics Letters, qualifies these findings and suggests that Bitcoin market efficiency has gradually improved as the market has matured and liquidity has increased

.

2.2. Arbitrage Pricing Theory (APT)

The Arbitrage Pricing Theory (APT), developed by Ross (1976) in an article published in the Journal of Economic Theory, provides a more flexible alternative to the Capital Asset Pricing Model (CAPM), which will be presented in the following section. Unlike the CAPM, the APT adopts a multifactor approach to explaining the returns of financial assets.

According to this model, an asset's return can be expressed as a linear function of several systematic factors:

:

$$R_i = \alpha_i + \beta_{1i}F_1 + \beta_{2i}F_2 + \cdots + \beta_{ki}F_k + \varepsilon_i$$

Where:

- R_i represents the return on asset \(i\).
- F_j for j=1…. k, represents the risk factors common to the market, such as economic growth, inflation, changes in interest rates, or the credit risk premium.
- β_{ji}measures the sensitivity of asset i to each of these factors.
- ε_i represents the asset-specific error term, reflecting its idiosyncratic risk.

One of the main advantages of the APT is that it does not rely on the restrictive assumptions of the CAPM, particularly regarding investors' preferences or the existence of a single market portfolio. However, it relies on one essential condition: the absence of arbitrage opportunities in financial markets.

The application of the APT to cryptocurrency markets is a relatively recent but rapidly developing area of research. In this context, Liu and Tsyvinski (2021), in a study published in the Review of Financial Studies, identify three risk factors specific to cryptocurrency markets:

1. Network size factor, measured by the number of active addresses or wallets.
2. Momentum factor, reflecting the persistence of recent price movements.
3. Value factor, capturing differences in valuation across crypto assets.

Their findings suggest that cryptocurrency returns follow a factor-based pattern that differs, at least partially, from the patterns observed in traditional stock markets.

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Section 3. Portfolio Management and the Capital Asset Pricing Model (CAPM)

Financial risk management occupies a central position in modern finance because it aims to preserve investment stability in the face of market uncertainty. It encompasses all the methods, tools, and practices designed to identify, assess, and control the various risks faced by investors and financial institutions.

Its objective is not to eliminate uncertainty entirely, but to ensure that the level of risk undertaken remains consistent with investors' strategic objectives and preferences. Risk

management therefore relies on a proactive approach: anticipating events that may affect financial performance and implementing appropriate mechanisms to limit their consequences.

The development of this discipline is closely linked to the pioneering work of Harry Markowitz (1952), whose modern portfolio theory introduced a quantitative approach to asset allocation based on the trade-off between expected return and risk. This contribution profoundly changed investment decision-making by providing a rigorous framework for portfolio diversification.

At the heart of this approach lies the measurement of risk, which involves quantifying exposure to losses through indicators that are reliable, interpretable, and useful for decision-making. In this regard, Artzner et al. (1999) made a major contribution by defining the properties that a risk measure must satisfy to be considered coherent. Through a set of axioms, they established a theoretical framework for distinguishing robust risk measures from indicators whose limitations reduce their relevance in certain situations.

This section presents the main foundations of modern portfolio theory. It begins by examining Markowitz's contributions, then considers the efficient frontier and its implications for optimal asset allocation. Finally, it discusses the main limitations of this traditional approach and introduces contemporary methods of portfolio construction and management developed to address the challenges of modern financial markets.

3.1. Markowitz's Modern Portfolio Theory

. Markowitz's theory is based on the idea that every investment decision must consider expected return and associated risk simultaneously. Expected return is measured by the mathematical expectation of returns, while risk is measured by variance or standard deviation.

The main contribution of this theory is to demonstrate that portfolio risk depends not only on the characteristics of individual assets but also on the relationships between their returns, measured through covariance. This approach led to the development of efficient portfolios and laid the foundations for several major models in modern finance, including the CAPM developed by Sharpe (1964) and Lintner (1965), as well as the multifactor models of Fama and French (1993).

3.1.1. Portfolio Return and Risk

Before presenting Markowitz's portfolio optimization problem, it is necessary to define its two fundamental components: expected portfolio return and portfolio risk, measured by variance. These concepts are first formulated for a general portfolio containing (N) assets and then illustrated in the particular case of a two-asset portfolio..

- ***Cas général à N actifs (notation matricielle)***

Consider a portfolio composed of (N) risky assets. Let : $w = (w_1, w_2, \ldots, w_n)'$ represent the column vector of portfolio weights, μ represent the vector of expected asset returns, et *represent the (N*N) variance-covariance matrix of asset returns*. The expected return and variance of the portfolio are expressed using these quantities. The key principle is that portfolio risk depends on both the individual variances of the assets and the covariances between their returns.

In other words, combining assets whose returns do not move perfectly together can reduce overall portfolio risk. This is the fundamental principle of diversification.

$$E(R_p) = w^T \mu = \sum_{i=1}^{N} w_i \, E(R_i)$$

$$\sigma_p^2 = w^T \Sigma w = \sum_{i=1}^{N} \sum_{j=1}^{N} w_i \, w_j \, \sigma_{ij}$$

Avec

$\sigma_{ij} = \sigma_i \, \sigma_j \, \rho_{ij}$ represents the covariance between asset i and j, while ρ_{ij} denotes their correlation coefficient. The matrix Σ contains the individual variances σ_i^2 along its main diagonal and the cross-covariances outside the diagonal. This reflects the fact that the overall risk of a portfolio depends not only on the individual volatility of each asset but also on the dependence structure among all the assets.

Finally, the portfolio is subject to the budget constraint $w^T 1 = 1$, where 1 is a column vector of N elements, all equal to 1

- ***Special case: Two assets***

The two-asset case, often used for educational purposes, follows directly from the general case by setting $N = 2$:

$$E(R_p) = w_A\, E(R_A) + w_B\, E(R_B)$$

$$\sigma_p^2 = w_A^2\, \sigma_A^2 + w_B^2\, \sigma_B^2 + 2 w_A\, w_B\, \sigma_A\, \sigma_B\, \rho_{AB}$$

$$\sigma_p = \sqrt{\sigma_p^2}$$

This simplified representation clearly highlights the role of the correlation coefficient ρAB :the lower the correlation, or even the more negative it is, the greater the potential for diversification to reduce the overall portfolio risk without reducing its expected return.

3.1.2. The Portfolio Optimization Problem

Once the portfolio's risk-return profile has been established, the rational investor will seek to optimize the allocation of assets. The objective is clear: to determine the weights (w) that minimize risk while achieving a specific target return. In practice, this problem is modeled as a constrained quadratic optimization problem.

$$min_w \;\; \sigma_p^2 = w^T \Sigma w$$

$$sous\ contraintes : \;\; w^T \mu = E(R_p^*) \quad et \quad w^T \mathbf{1} = 1$$

In practice, this model is based on two fundamental constraints. The first requires the portfolio to achieve the target expected return $E(R_p^*)$ set by the investor. The second is the standard budget constraint: the entire capital must be invested.

Interestingly, the logic of this problem can be reversed through what is known as its dual formulation. Instead of minimizing risk for a given target return, the objective becomes maximizing return for a given level of risk that the investor is willing not to exceed.

Finally, depending on the chosen strategy, an additional constraint may be imposed, such as prohibiting short selling. Mathematically, this means that all portfolio weights must be non-negative, that is ($w_i \geq 0$)

3.1.3. Determining Optimal Weights

The optimization problem is solved using the Lagrangian method, which incorporates the two constraints through the multipliers $\lambda 1$ and $\lambda 2$:

$$L(w, \lambda_1, \lambda_2) = w^T \Sigma w - \lambda_1 \left(w^T \mu - E(R_p^*)\right) - \lambda_2 (w^T \mathbf{1} - 1)$$

The first-order conditions ($\partial L/\partial w = 0$) lead to the following expression for the optimal portfolio weights:

$$w^* = \Sigma^{-1}(\lambda_1 \mu + \lambda_2 1)$$

By substituting this expression into the two constraints and defining the scalar constants $A = \mu^T \Sigma^{-1} \mu$, $B = \mu^T \Sigma^{-1} 1$ and $C = 1^T \Sigma^{-1} 1$ (Merton, 1972), we obtain the explicit solution for the multipliers and, subsequently, the optimal portfolio weights as a function of the target return:

$$w^* = \left[\frac{C\, E(R_p^*) - B}{AC - B^2}\right] \Sigma^{-1} \mu + \left[\frac{A - B\, E(R_p^*)}{AC - B^2}\right] \Sigma^{-1} \mathbf{1}$$

This formula highlights a fundamental property: the optimal portfolio can be constructed as a simple linear combination of two benchmark portfolios. This is the core principle of the two-fund separation theorem. In other words, to replicate any efficient portfolio, an investor needs only two basic building blocks: a combination of two efficient portfolios.

3.1.4. The Efficient Frontier

By substituting the optimal weights w*into the portfolio variance formula, we obtain the equation of the efficient frontier. This equation is fundamental because it directly establishes the relationship between the minimum attainable risk and the expected return level.

$$\sigma_p^2 = \frac{C\, E(R^2) - 2B\, E(R) + A}{AC - B^2}$$

This relationship takes the form of a parabola in variance-return space and a hyperbola in the standard deviation-return plane, which is commonly used for graphical representations. It thus embodies the founding principle of Markowitz's theory: for each target return level, there is a unique portfolio that achieves the strictly minimum variance.

3.1.5. The Efficient Frontier Curve

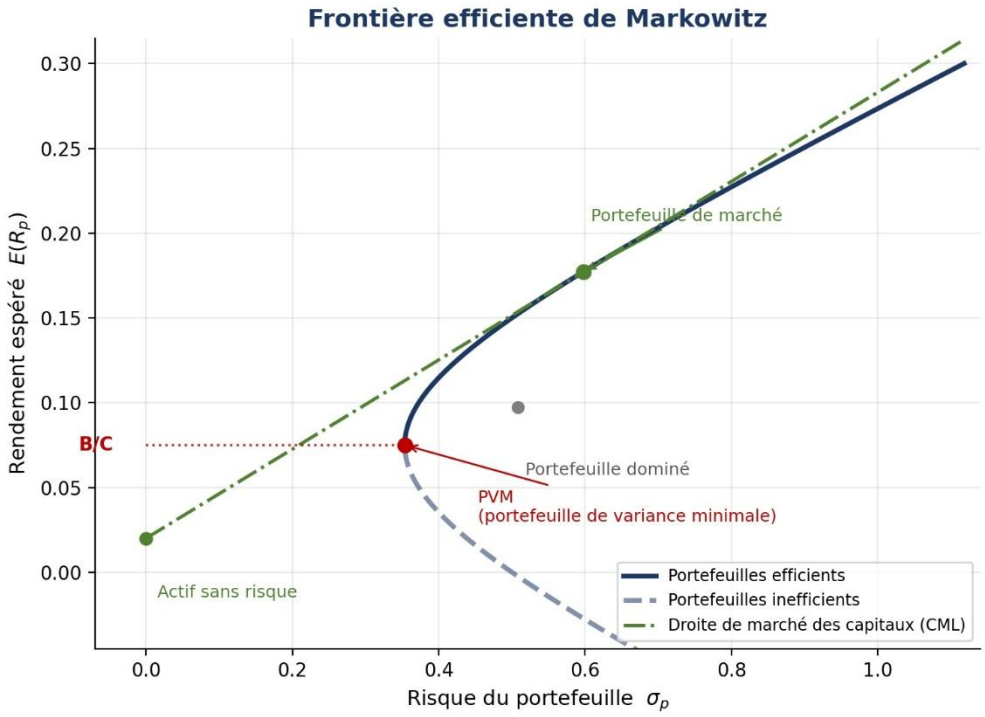


Figure 1. Markowitz Efficient Frontier

Figure 1 illustrates the appearance of the efficient frontier in risk-return space. It shows a convex curve whose leftmost point corresponds to the minimum-variance portfolio, obtained when (E(Rp) = B/C).

This curve naturally divides into two parts. The upper branch represents the efficient frontier: it contains the best portfolios, those offering the highest return for each level of risk. The lower branch, by contrast, consists of inefficient portfolios. Why? Because they are systematically dominated: for the same level of risk, a better portfolio with a higher return can always be found on the upper branch.

In other words, any portfolio lying below this frontier is suboptimal. There is always a better combination of assets available for the same level of risk.

Later, this theory was extended by the Capital Asset Pricing Model (CAPM) developed by Sharpe (1964) and Lintner (1965). Their key idea was to introduce a risk-free asset and the market portfolio as a benchmark. This made it possible to establish a linear relationship—the Capital Market Line—between an asset's expected return and its systematic risk, measured by the well-known beta coefficient.

Even today, the efficient frontier remains an essential tool for asset managers.

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3.2. The Capital Asset Pricing Model (CAPM)

The Capital Asset Pricing Model (CAPM) is one of the most widely used equilibrium models in finance for explaining the relationship between risk and the expected return of an asset. Developed independently by Sharpe (1964), Lintner (1965), and Mossin (1966), this model extends Markowitz's theory by introducing the concept of a risk-free asset and establishing equilibrium conditions in a competitive market.

The central equation of the CAPM corresponds to the Security Market Line (SML):

$$E(R_i) = R_f + \beta_i\left[E(R_m) - R_f\right]$$

Where $E(R_i)$ denotes the expected return of asset i, R_f represents the risk-free rate (generally approximated by short-term Treasury bills), $E(R_m)$ is the expected return of the market portfolio, and $\left[E(R_m) - R_f\right]$ is the market risk premium. The coefficient β_i measures the asset's systematic risk, that is, the component of risk that cannot be diversified away.

Beta is defined as follows:

$$\beta_i = \frac{Cov(R_i, R_m)}{Var(R_m)} = \rho_{im}\,\frac{\sigma_i}{\sigma_m}$$

It measures the sensitivity of an asset's return to market fluctuations. A beta greater than 1 indicates that the asset amplifies market movements and is therefore considered riskier. A beta lower than 1 indicates lower sensitivity to market fluctuations, characterizing a more defensive asset. Finally, a negative beta means that the asset moves in the opposite direction to the market, potentially giving it a hedging role. In the context of cryptocurrencies, the stability of beta relative to traditional financial markets has become a major issue in recent research on financial connectedness.

3.3. Extensions of the CAPM and Multifactor Models

Empirical tests of the CAPM quickly highlighted certain limitations, particularly the fact that the market factor alone is insufficient to explain differences in returns across assets. In this context, Fama and French (1993), in a study published in the Journal of Financial Economics, proposed a three-factor model that incorporates, in addition to the market factor, a size factor (SMB: Small Minus Big), which captures the performance difference between small- and large-capitalization stocks, and a value factor (HML: High Minus Low), which reflects the return difference between stocks with high and low book-to-market ratios.

The model is expressed as follows:

$$E(R_i) - R_f = \alpha_i + \beta_i\left[E(R_m) - R_f\right] + s_i\, SMB + h_i\, HML$$

Subsequently, Carhart (1997) extended this framework by adding a fourth factor, momentum (WML: Winners Minus Losers), which reflects the tendency of assets that have recently outperformed to continue outperforming in the short term. Finally, Fama and French (2015) proposed a five-factor model by introducing two additional variables: profitability (RMW: Robust Minus Weak) and investment (CMA: Conservative Minus Aggressive), thereby providing a more refined explanation of financial asset returns.

3.4. The Sharpe Ratio and Portfolio Optimization

The Sharpe ratio, introduced by William Sharpe (1966) in the Journal of Business, is one of the most widely used measures for evaluating a portfolio's risk-adjusted performance. It measures the excess return earned per unit of total risk borne by the investor.

The ratio is expressed as follows:

$$SR = \frac{E\left(R_p\right) - R_f}{\sigma_p}$$

Where $E(R_p)$ represents the expected portfolio return, R_f is the risk-free rate, and σ_p is the standard deviation of portfolio returns, which measures overall portfolio risk. The higher the Sharpe ratio, the more satisfactory the portfolio's performance is considered relative to the risk taken.

When integrating cryptocurrencies into a portfolio of traditional assets, this ratio is particularly relevant for assessing whether their inclusion improves the overall portfolio's risk-adjusted performance, despite their high individual volatility.

From this perspective, several recent studies have examined the impact of cryptocurrencies on portfolio performance. Platanakis and Urquhart (2020), in a study published in the European Journal of Finance, show that the optimal allocation including Bitcoin depends heavily on the assumptions underlying the optimization model and the period analyzed. Their findings therefore highlight the need to adopt robust approaches to reliably assess the contribution of cryptocurrencies to overall portfolio performance.

Conclusion

Ultimately, this first chapter has enabled us to establish the theoretical framework of our research topic. We began with the fundamentals of traditional financial markets before exploring the more volatile and technologically driven ecosystem of cryptocurrencies. We also revisited the major models of modern finance, from Markowitz's efficient frontier to the Capital Asset Pricing Model (CAPM), which remind us that diversification is the only "free lunch" in finance.

However, the practical realities of financial markets and our literature review have revealed a major challenge: traditional models assume relatively stable markets and consistent relationships between assets. Yet, as we have seen, cryptocurrencies challenge these assumptions because of their volatility and sensitivity to investor sentiment. This is where the concept of financial connectedness comes into play, allowing us to measure how crises and shocks propagate from one market to another.

Nevertheless, measuring these relationships is not enough; we must also anticipate them. Given the limitations of traditional econometric tools, we will turn to artificial intelligence. The following chapter (or empirical section) will therefore introduce our methodological toolkit, explaining how to combine dynamic connectedness analysis using the Quantile Vector Autoregression (QVAR) model with the power of Machine Learning, particularly Random Forest and Isolation Forest, in order not only to forecast risks but also to construct more resilient and higher-performing portfolios.

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CHAPTER 2: Extreme Connectedness Analysis, Artificial Intelligence-Based Prediction, and Optimal Portfolio Strategies

Introduction

This chapter provides the theoretical and methodological foundation of this research. First, it aims to present the main contributions in the literature concerning the integration of cryptocurrencies into financial markets, their role in portfolio diversification, the mechanisms through which financial shocks are transmitted, and recent modeling and forecasting approaches based on econometric techniques and artificial intelligence. This literature review highlights existing scientific advances while identifying the limitations that the present study seeks to address.

Second, this chapter presents the adopted methodological framework. It introduces the foundations of the Quantile Vector Autoregression (QVAR) model, which is used to measure connectedness between financial markets under different market regimes, as well as the Machine Learning models employed to predict changes in connectedness and detect periods of extreme stress. The different portfolio management strategies (MVP, MCP, and MCoP), together with the performance and risk indicators used to evaluate them, are also described in detail.

Finally, the chapter presents the characteristics of the dataset through a descriptive analysis of returns, followed by an initial examination of the structure of interdependencies between markets using static and dynamic connectedness measures. This step highlights the mechanisms through which shocks are transmitted between financial assets and establishes the empirical foundations for the predictive analyses and portfolio allocation strategy evaluations developed in the following chapters.

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Section 1. Literature Review

1.1. Integration of Cryptocurrencies into Financial Markets

Andrew Urquhart (2016) demonstrates that the Bitcoin market does not satisfy the weak-form Efficient Market Hypothesis over the entire period examined. Nevertheless, the

empirical results reveal that, during the second subperiod, certain tests suggest a movement toward greater randomness in returns, indicating a gradual improvement in the market's informational efficiency. Despite this trend, the author concludes that Bitcoin remains generally inefficient. This inefficiency is primarily explained by the relatively recent emergence of this asset, which can be compared to an emerging market in its development phase. The study also emphasizes that an increase in the number of market participants, together with more intensive trading and analytical activity, could gradually promote greater market efficiency over time. Finally, the author recommends further research into the temporal dynamics of Bitcoin's efficiency and comparisons with emerging markets and other alternative asset classes.

Corbet et al. (2020) examine the behavior of cryptocurrencies during the COVID-19 health crisis to assess their potential role as safe-haven assets during periods of severe financial instability. Their analysis takes place in a context in which traditional flight-to-safety mechanisms were being questioned, particularly in energy markets, which were disrupted by the economic and geopolitical tensions associated with the pandemic. The authors use data on returns and trading volumes for major cryptocurrencies traded on the Binance platform between January 2019 and March 2020. They complement this approach with a market sentiment analysis based on more than 397,000 tweets relating to COVID-19 and cryptocurrencies. Natural Language Processing (NLP) techniques are employed to assess the polarity and subjectivity of the opinions expressed. The study also uses a GARCH(1,1) model to analyze the impact of pandemic-related sentiment on cryptocurrency return dynamics.

The results show that major cryptocurrencies experienced increases in returns and trading volumes during the crisis, suggesting their potential role as stores of value and safe-haven assets for investors. Furthermore, the authors identify a strong sensitivity of cryptocurrency returns to negative sentiment associated with COVID-19, confirming the significant influence of social media on price formation. Cryptocurrencies also exhibit characteristics similar to those of precious metals during periods of crisis, reinforcing their potential value for portfolio diversification. However, despite these findings, the study highlights the lack of a clear regulatory framework and the risks of market manipulation as important limitations. The authors conclude that cryptocurrencies have gained credibility as financial assets that may play a safe-haven role during periods of extreme market stress.

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1.2. Portfolio Diversification and Spillovers Between Cryptocurrencies and Traditional Markets

Seongcheol Paeng et al. (2024) demonstrate that incorporating stablecoins into an investment portfolio can generate significant diversification benefits and reduce risk. The authors emphasize that these assets exhibit low or even negative correlations with other financial asset classes, thereby helping to improve risk-adjusted performance, particularly through the stable returns offered by certain decentralized finance (DeFi) platforms. The study also identifies causal relationships and transmission effects between stablecoins, cryptocurrencies, and traditional financial markets. According to the authors, the high volatility of cryptocurrencies and their substantial transaction costs often encourage investors to use stablecoins first before investing in assets such as Bitcoin and Ethereum.

The results also indicate that the performance of the S&P 500 has a delayed influence on stablecoins, which subsequently affect major cryptocurrencies. Following the COVID-19 crisis, causal relationships and spillover effects between stock markets, stablecoins—particularly USDT and USDC—and other cryptocurrencies intensified, reflecting growing interconnectedness between traditional financial markets and digital asset markets.

Bouri et al. (2017) analyze Bitcoin's diversification, hedging, and safe-haven properties in relation to several international asset classes, including global equities, bonds, oil, gold, commodities, and the US dollar index. Using daily and weekly data and employing a Dynamic Conditional Correlation (DCC) model, the authors conclude that Bitcoin generally constitutes an effective diversification instrument in most of the configurations examined. They also show that Bitcoin can, under certain specific conditions, act as a hedging instrument and a safe-haven asset, although these properties depend heavily on the time horizon considered. However, the authors emphasize that Bitcoin's relatively low liquidity and high volatility limit the stability of its diversification and protection benefits over time. They therefore stress the evolving nature of these properties and recommend further research into their temporal dynamics.

Bhuiyan et al. (2021), Ghabri et al. (2020), and Huynh et al. (2020a, 2020b) highlight Bitcoin's role as a hedging instrument against various financial risks. Several studies show that, like gold, Bitcoin can provide a hedge against developed markets and maintain this property under both normal market conditions and periods of financial stress (Jeribi and Ghorbel, 2021; Zeng et al., 2020; Chemkha et al., 2021). Vo et al. (2021) further emphasize that Bitcoin has

gradually evolved from an asset primarily associated with speculation into an investment instrument that is increasingly sensitive to macroeconomic factors.

Empirical literature also indicates that Bitcoin can protect investors against increased asset volatility across several major economies, including the United States, Germany, France, China, Canada, Russia, the United Kingdom, and Japan (Mokni, 2021). Other studies confirm its hedging role in relation to sectoral portfolios, bonds, the oil market, and certain stock markets, particularly those of Finland, the Netherlands, and the United States (Akhtaruzzaman et al., 2020; Ghorbel and Jeribi, 2021a; Urom et al., 2020).

Furthermore, Bitcoin also exhibits hedging properties across different categories of investments, including sustainable, Islamic, and traditional investments, as well as certain agricultural and metal commodities (Disli et al., 2021; Fakhfekh et al., 2021; Naeem et al., 2021a). Kinkyo (2020) also shows that Bitcoin can act as a hedge against several Asian fiat currencies, outperforming gold and oil over medium- and long-term horizons. Finally, González et al. (2021) demonstrate that Bitcoin can also serve as a hedge against gold and assets strongly correlated with gold, particularly during the COVID-19 pandemic, owing to its asymmetric responses to changes in gold returns.

1.3. Forecasting and Modeling

Abdul Moiz et al. (2026) examine the application of conventional econometric models, deep learning architectures, and hybrid approaches to volatility forecasting in cryptocurrency markets. Using daily data covering ten major cryptocurrencies over the 2020–2025 period, the authors compare the performance of GARCH models, deep learning models, and hybrid approaches using an evaluation framework based on 30-day realized volatility. The results indicate that traditional GARCH models remain effective for relatively stable assets, whereas deep learning models improve forecasting accuracy through their ability to capture nonlinear market dynamics.

The study particularly highlights the superiority of hybrid models, especially those combining GARCH structures with architectures such as Time Delay Neural Networks (TDNN) and Gated Recurrent Units (GRU). These models produce the most accurate and statistically robust volatility forecasts. The authors conclude that hybrid approaches significantly improve the modeling and management of cryptocurrency-related risks.

Mari and Ngunyi (2021) propose a GARCH-EVT model incorporating heavy tails to improve cryptocurrency volatility forecasting, particularly for Bitcoin. By combining Extreme Value Theory (EVT) with GARCH models, the authors seek to better represent the extreme events characteristic of these markets. However, Micu and Dumitrescu (2022) argue that standard GARCH models remain insufficient to fully capture cryptocurrency volatility dynamics because of their considerable instability and specific characteristics. Similarly, Catania and Grassi (2022) demonstrate that traditional GARCH models have limited performance when estimating cryptocurrency returns, whereas score-driven models provide more accurate results. These studies highlight the need to develop approaches better suited to the complexity of digital asset markets.

Continuing along this line of research, Aras (2021) proposes a stacking ensemble approach that combines several base models with a Support Vector Machine (SVM)-based meta-learner to forecast Bitcoin volatility. The results show that hybrid models incorporating an appropriate selection of explanatory variables can significantly improve forecasting quality. Similarly, Alghamdi et al. (2022) employ Long Short-Term Memory (LSTM) models combined with investor sentiment measures to explain cryptocurrency volatility, thereby expanding the application of deep learning techniques in this field.

Recent studies confirm this trend. Garcia-Medina et al. (2024) develop a hybrid LSTM-GARCH model that combines volatility measures derived from GARCH models with the learning capabilities of LSTM networks, improving forecasting accuracy. Meanwhile, Tripathy et al. (2025) demonstrate that a multivariate bidirectional LSTM (Bi-LSTM) model outperforms GARCH and TGARCH models in Bitcoin volatility forecasting, confirming the superior performance of multivariate deep learning architectures. Finally, Iqbal et al. (2025) combine the Nonlinear GARCH (NGARCH) model with several Machine Learning algorithms, including SVM, K-Nearest Neighbors (KNN), and Artificial Neural Networks (ANN). Their results indicate that the hybrid NGARCH-KNN approach achieves the best performance in directional volatility forecasting.

Section 2. Methodology and Modeling: QVAR and Machine Learning

2.1. The QVAR Model and the Measurement of Connectedness Across Quantiles

Analyzing the transmission of shocks between financial markets requires econometric tools capable of capturing the complexity of nonlinear dynamics and the asymmetric behavior of investors. Our methodology is built around two innovative pillars: (1) measuring extreme connectedness using the Quantile Vector Autoregression (QVAR) model, and (2) predicting changes in connectedness and detecting anomalies through Machine Learning (Artificial Intelligence).

2.1.1. Rationale for the QVAR Approach

The traditional connectedness approach developed by Diebold and Yilmaz (2012, 2014) relies on a Vector Autoregression (VAR) model estimated using Ordinary Least Squares (OLS). This method has a fundamental limitation: it models only the conditional mean of the return distribution. However, financial markets are characterized by fat-tailed distributions and pronounced asymmetry. Risk managers are particularly concerned about extreme events, such as market crashes or periods of euphoria, which occur in the tails of the distribution.

To overcome this limitation, we employ the QVAR approach (Quantile Vector Autoregression), proposed by Ando et al. (2022) and adapted to connectedness analysis by Chatziantoniou et al. (2021). QVAR makes it possible to estimate model parameters at different quantiles $\tau \in (0,1)$ of the conditional distribution:

- Le quantile $\tau = 0.5$ represents normal market conditions (the median).
- Le quantile $\tau = 0.95$ captures extreme positive shocks (bull markets or euphoria).
- Le quantile $\tau = 0.05$ captures extreme negative shocks (bear markets or crashes).

This approach is crucial to our study because it allows us to demonstrate that connectedness is not static: it can increase sharply during crises due to contagion effects and undergo structural changes during major geopolitical shocks, such as the US–Iran conflict.

2.1.2. Mathematical Specification of QVAR

Let Y_t vector containing the returns of our N= 8 financial assets at time t. QVAR(p) model is expressed as follows:

$$y_t = \mu(\tau) + \sum_{\mathrm{j}}^{\mathrm{p}} \Phi_{\mathrm{j}}(\tau) y_{t-j} + u_t(\tau) = \mu(\tau) + \sum_{i=0}^{\infty} \Omega_i(\tau) u_{t-i} \ \tau \in (0,1)$$

Où :

$\mu(\tau)$: is the vector of intercepts specific to quantile τ

$\Phi_i(\tau)$: represents the N * N autoregressive coefficient matrices at quantile τ

$\epsilon_t(\tau)$: is the vector of error terms, subject to the quantile condition: $Q_\tau(\epsilon_t(\tau)|Y_{t-1}, \dots) = 0$

Unlike the standard VAR model, the parameters Φ_i vary with the quantile τ. This makes it possible to capture how network dynamics change when the system experiences shocks of different magnitudes.

2.1.3. Quantile-Based Generalized Forecast Error Variance Decomposition (GFEVD)

To measure connectedness, we use the Generalized Forecast Error Variance Decomposition (GFEVD), adapted to different quantiles. The Wold representation of the QVAR model is:

$$Y_t = \sum_{s=0}^{\infty} \Phi_s(\tau)\epsilon_{t-s}(\tau) \quad (1)$$

La part de la variance de l'erreur de prévision à l'horizon H de la variable i due aux chocs de la variable j au quantile τ est donnée par :

The proportion of the forecast error variance of variable i, at horizon H, attributable to shocks in variable j at quantile τ is given by:

$$\widetilde{\theta_{ij}^{q}}(H,\tau) = \frac{\sigma_{jj}^{-1}(\tau)\sum_{s=0}^{H-1}\left(e_i'\Phi_s(\tau)\Sigma(\tau)e_j\right)^2}{\sum_{s=0}^{H-1}(e_i'\Phi_s(\tau)\Sigma(\tau)\Phi_s(\tau)'e_i)} \quad (2)$$

En normalisant cette matrice pour que la somme des lignes soit égale à 1, on obtient la matrice de connectivité $\widetilde{\theta_{ij}^{q}}(H,\tau)$. À partir de cette matrice, nous extrayons les mesures agrégées de la topologie du réseau :

Where $\sum(\tau)$ is the covariance matrix of the error terms at quantile τ_i, and e_i are selection vectors corresponding to the i-th and j-th variables, respectively.

By normalizing this matrix so that the sum of each row equals 1, we obtain the connectedness matrix $\widetilde{\theta_{ij}^{q}}(H,\tau)$. From this matrix, we derive aggregate measures describing the network's structure:

- Directional Connectedness “FROM” (Receiver) :This measures the proportion of the variance of asset i explained by shocks originating from all other assets.

$$FROM_i^q(\tau) = \sum_{j=1, j\neq i}^{N} \overline{\theta_{ij}^{q}}(H,\tau) \times 100 \quad (3)$$

- Directional Connectedness "TO" (Transmitter) This measures the shocks transmitted by asset i to all other assets.

$$TO_i^q(\tau) = \sum_{j=1, j\neq i}^{N} \overline{\theta_{ji}^q}(H,\tau) \times 100 \quad (4)$$

- Net Connectedness (NET): This represents the difference between the shocks transmitted and those received by an asset.

- Total Connectedness index (TCI) : This provides an aggregate measure of systemic risk across the financial network.

$$TCI^q(\tau) = \frac{\sum_{i=1, i\neq j}^{N} \overline{\theta_{ij}^q}(H,\tau)}{\sum_{i,j=1}^{N} \overline{\theta_{ij}^q}(H,\tau)} \times 100 (6)$$

By applying a rolling window, we obtain a daily time series of $TCI^q(\tau)$, which will serve as the target variable for our Machine Learning model.

2.2. The Predictive Approach Based on Artificial Intelligence (Machine Learning)

2.2.1. Rationale for Using Machine Learning

Traditional literature (e.g., Baruník and Křehlík, 2018) often uses frequency decomposition to analyze short-, medium-, and long-term connectedness. However, this approach is purely descriptive (ex post) and is difficult for a risk manager to use for forecasting purposes (ex-ante).

To introduce an innovative approach, we replace this decomposition with a nonlinear predictive approach. The TCI time series extracted from the QVAR model exhibits nonlinearities, structural breaks, and complex dependencies that classical models (such as ARIMA) cannot capture. We use decision-tree-based ensemble algorithms (Random Forest Improved, XGBoost, and LightGBM) to forecast the future TCI. In addition, we use Isolation Forest to detect structural anomalies (crises) objectively, without making any underlying distributional assumptions.

2.2.2. Single-Layer Prediction Model and Equations

The Random Forest model was trained to estimate the Total Connectedness Index (TCI) using the eight financial variables (WTI, GOLD, SSE, STX600, SP500, DXY, BTC, and ETH). Each decision tree is built from a bootstrap sample of the data and uses a random selection of explanatory variables at each node. The final prediction is the average of the individual tree

predictions, which reduces variance and improves the model's robustness and generalization ability.

Let the training dataset be:

$$D = \{(X_i, TCI_i)\}_{\{i=1\}}^{\{N\}}$$

Où

$X_i = (WTI_i, GOLD_i, SSE_i, STX600_i, SP500_i, DXY_i, BTC_i, ETH_i)$ represents the vector of returns for the eight financial variables on day i

TCI_i is the Total Connectedness Index

N is the number of observations.

Random Forest constructs B independent decision trees from bootstrap samples:

$$T = \{T_1, T_2, \dots, T_B\}$$

The Random Forest prediction is obtained by averaging the predictions of the B trees:

$$\widehat{TCI} = \frac{1}{B}\sum_{b=1}^{B} T_b(X) \quad (7)$$

The final model is trained by learning the function :

$$\widehat{f_{RF}}(X) = \frac{1}{B}\sum_{b=1}^{B} T_b(X) \quad (8)$$

To evaluate the predictive performance of the one-layer and two-layer models, we use the following metrics:

Root Mean Squared Error (RMSE): penalizes large forecasting errors

$$RMSE = \sqrt{\frac{1}{T}\sum_{t=1}^{T}\left(TCI_t - \widehat{TCI_t}\right)^2} \quad (9)$$

Mean Absolute Error (MAE): measures the absolute magnitude of the errors

$$MAE = \frac{1}{T}\sum_{t=1}^{T}\left|TCI_t - \widehat{TCI_t}\right| \quad (10)$$

Coefficient of Determination (R^2): the proportion of TCI variance explained by the model

$$R^2 = 1 - \frac{\sum_{t=1}^{T}(TCI_t - \widehat{TCI_t})^2}{\sum_{t=1}^{T}\left(TCI_t - \overline{TCI}\right)^2} \quad (11)$$

2.2.3. Anomaly Detection Using Random Forest (First Layer)

To identify panic days or "Black Swan" events (such as spikes associated with the US–Iran conflict), we use the Isolation Forest algorithm (Liu et al., 2008). Unlike conventional clustering methods, which model normal observations, Isolation Forest directly isolates anomalies.

The underlying principle is that anomalies are "few and different," making them easier to isolate in a decision tree. The anomaly score s(x, n) for an observation x in a sample of size n is calculated as follows

$$s(x,n) = 2^{-\frac{E(h(x))}{c(n)}} \quad (12)$$

Où :

E(h(x)) is the average path length in the tree required to isolate observation x

C(n) is the normalization factor, given by $c(n) = 2H(n-1) - \frac{2(n-1)}{n}$

et H(i) is the harmonic number.

If $s(x,n)$ is close to 1, the observation is a clear anomaly (an extreme crisis). If it is close to 0.5, the market is in a normal state. This enables us to define market regimes mathematically.

$$Regime_t = \begin{cases} Crisis & if \quad s(x,n) > Anomaly\ threshold \\ Normal & otherwise \end{cases}$$

2.3. Refinement and Robust Forecasting with Two-Layer Models (RF Improved, XGBoost, LightGBM)

Although standard Random Forest is robust, it may lack the precision (due to bias) required to capture extreme TCI spikes during sudden market crashes (e.g., the US–Iran conflict or COVID-19). The second layer of our model introduces advanced algorithms to refine forecasts, minimize residual errors, and compare predictive performance. We trained three advanced models.

2.3.1. Improved Random Forest

This is a Random Forest model whose hyperparameters (maximum depth, number of trees, and minimum samples per leaf) have been rigorously optimized using Grid Search and Time Series Cross-Validation to adapt specifically to the TCI's heavy-tailed distribution.

2.3.2. XGBoost (Extreme Gradient Boosting)

Unlike bagging, XGBoost uses sequential boosting. Each new tree is built to correct the residual errors of the preceding trees. Its main innovation is the addition of L2 regularization, which penalizes extreme weights and is crucial for avoiding false crisis signals. At iteration t, XGBoost minimizes an objective function Obj^(t), which combines a differentiable loss function L (the TCI prediction error) with a regularization term Ω(f_t) that controls model complexity:

:

$$\text{Obj}^{(t)} = \sum_{i=1}^{n} L\left(y_i, \widehat{y_i^{(t-1)}} + f_t(x_i)\right) + \Omega(f_t) \quad (13)$$

2.3.3. LightGBM (Light Gradient Boosting Machine)

Designed for speed and efficiency with large time series, LightGBM uses leaf-wise tree growth and the Gradient-based One-Side Sampling (GOSS) technique. GOSS retains observations with large gradients (including TCI spikes during crises) while removing observations with small gradients.

To determine the best split at a node, LightGBM calculates the gain ΔL. Let I be the set of observations in the parent node, I_L and I_R the sets in the child nodes, and g_i and $h_i$the gradient and Hessian of the loss function, respectively:

$$\Delta L = \frac{1}{2}\left[\frac{\left(\sum_{i\in I_L} g_i\right)^2}{\sum_{i\in I_L} h_i+\lambda} + \frac{\left(\sum_{i\in I_R} g_i\right)^2}{\sum_{i\in I_R} h_i+\lambda} - \frac{(\sum_{i\in I} g_i)^2}{\sum_{i\in I} h_i+\lambda}\right] - \gamma \quad (14)$$

The combined use of these three models provides strong predictive robustness. RF Improved offers stability in the presence of noise; XGBoost can improve precision through regularization, reducing false signals; and LightGBM is particularly effective at rapidly detecting regime changes, including sudden TCI spikes during geopolitical crashes. Comparing these models helps identify the most suitable early-warning system for a risk manager.

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2.4. Portfolio Management Strategies

Measuring connectedness and predicting shocks using artificial intelligence are only the first steps. The ultimate objective of this modelling for an asset manager is the optimal allocation of capital. With this in mind, we compare three multivariate strategies based on fundamentally different risk concepts: variance (MVP), correlation (MCP), and network connectedness (MCoP).

2.4.1. Minimum Variance Portfolio (MVP) – Markowitz (1952)

The Minimum Variance Portfolio (MVP), introduced by Markowitz (1952), is a cornerstone of modern portfolio theory. This strategy seeks to construct a portfolio with the lowest possible volatility by minimizing return variance while exploiting the diversification benefits offered by financial assets. The optimal weights are determined from the conditional variance–covariance matrix of asset returns and are expressed as follows:

$$W_{Ht} = \frac{H_t^{-1} I}{I' H_t^{-1} I} \quad (15)$$

Where W_{Ht} is an m × 1 vector of optimal portfolio weights, H_t is the conditional variance–covariance matrix of returns for the m assets at time t (with dimensions $m \times m$), et I I is an m-dimensional column vector whose elements are all equal to 1 (the unit vector). In all three

weighting formulas presented in this section, I impose the budget constraint that the portfolio weights sum to 1.

2.4.2. Minimum Correlation Portfolio (MCP) – Christoffersen et al. (2014)

To go beyond an approach based solely on covariance, Christoffersen et al. (2014) propose the Minimum Correlation Portfolio (MCP). This strategy relies on the conditional correlation matrix of the assets rather than their covariance matrix, thereby reducing the influence of scale differences between assets and improving portfolio diversification. The conditional correlation matrix is defined as :

$$R_t = diag(H_t)^{-0.5} H_t \, diag(H_t)^{-0.5} \quad (16)$$

Où *R*t is the conditional correlation matrix of returns, with dimensions m × m. The weights of the minimum-correlation portfolio are then obtained as follows:

$$W_{Rt} = \frac{R_t^{-1} I}{I' R_t^{-1} I} \quad (17)$$

2.4.3. Minimum Connectedness Portfolio (MCoP) – Broadstock et al. (2022)

More recently, Broadstock et al. (2022) proposed the Minimum Connectedness Portfolio, based on pairwise connectedness indices between financial assets. Unlike traditional approaches based on volatility or correlation, this method aims to limit shock-transmission mechanisms between assets by favoring those with lower connectedness levels. By reducing interdependencies within the portfolio, this strategy can improve resilience to systemic risk and episodes of financial contagion. Assets that exert little influence on other markets and are themselves only weakly affected by external shocks receive larger weights. The optimal weights are obtained from:

$$W_t = \frac{PCI_t^{-1} I}{I' PCI_t^{-1} I} (18)$$

PCIt denotes the pairwise connectedness index matrix (for returns or volatilities), with dimensions m × m, and I is the unit vector. This formulation therefore has exactly the same mathematical structure as the MVP and MCP portfolios; the only difference is the risk matrix used as input: variance–covariance for the MVP, correlation for the MCP, and connectedness for the MCoP.

2.5. Performance and Extreme Risk Measures

To assess the empirical superiority of the MCoP strategy relative to MVP and MCP, we use risk-adjusted performance metrics and tail-risk measures.

2.5.1. Sharpe Ratio (SR)

The Sharpe Ratio evaluates a portfolio's performance in relation to the total risk (volatility) it generates. It measures the risk premium earned per unit of volatility.

$$SR = \frac{R_p - R_f}{\sigma_p} \quad (19)$$

Here R_p is the portfolio's annualized return, R_f is the risk-free rate, and σ_p is the standard deviation of portfolio returns. A higher SR indicates that a strategy (such as MCoP) generates better returns for a given level of fluctuation, demonstrating the effectiveness of structural de-coupling.

2.5.2. Value at Risk (VaR) at 95%

Value at Risk is a standard measure for quantifying extreme-loss risk. It represents the maximum expected loss over a given horizon at a 95% confidence level.

$$VaR_{95\backslash\%} = \inf\{l \in R : P(L > l) \leq 0{,}05\} \quad (20)$$

Here, L represents the distribution of portfolio losses. VaR does not indicate the magnitude of losses beyond the 95% threshold, which is problematic during "Black Swan" events such as market crashes and wars.

2.5.3. Expected Shortfall (ES) at 95%

To capture the severity of crashes (such as the panic spike during the US–Iran conflict), we use Expected Shortfall (or Conditional VaR). ES measures the average loss conditional on the loss exceeding the VaR threshold. It is a coherent risk measure (Artzner et al., 1999).

$$ES_{95\backslash\%} = E\left[-R_p \mid R_p < -VaR_{95\backslash\%}\right] = \frac{1}{0.05}\int_0^{0.05} VaR_q \; dq \quad (21)$$

ES helps determine whether MCoP genuinely protects investors' capital against systemic market collapses (the worst 5% of days in the sample), where Markowitz's MVP typically exhibits its greatest weaknesses.

2.6. Data Processing

Before undertaking any econometric modeling, particularly the estimation of the QVAR model, it is essential to rigorously transform the raw time series. Financial series, such as daily asset prices, are characterized by stochastic trends and time-varying variance. Statistically, they are non-stationary, meaning that they contain a unit root.

Estimating a vector autoregressive model using non-stationary series could lead to biased results and spurious correlations, where by the model identifies relationships between assets that do not actually exist simply because they share a common time trend. To address this issue and ensure the stationarity of our series, we apply the first difference of logarithmic prices, allowing us to move from price levels to continuously compounded financial returns.

The daily return of each asset is calculated using the following formula:

$$r_t = \ln(P_t) - \ln(P_{t-1}) = \ln\left(\frac{P_t}{P_{t-1}}\right) \quad (22)$$

Où : r_t represents the logarithmic return of asset i on day t. P_t s the closing price of asset i on day t.P_{t-1} is the closing price of the asset on the previous day. ln denotes the natural logarithm.

2.6.1. Visual Analysis and Interpretation of Return Dynamics

The observation of daily return curves (Figure 2) over our study period (2020–June 2026) highlights the stylized facts of financial time series and directly reflects the impact of the three major systemic shocks in our sample: the COVID-19 pandemic, the Russia–Ukraine war, and the recent US–Iran conflict.

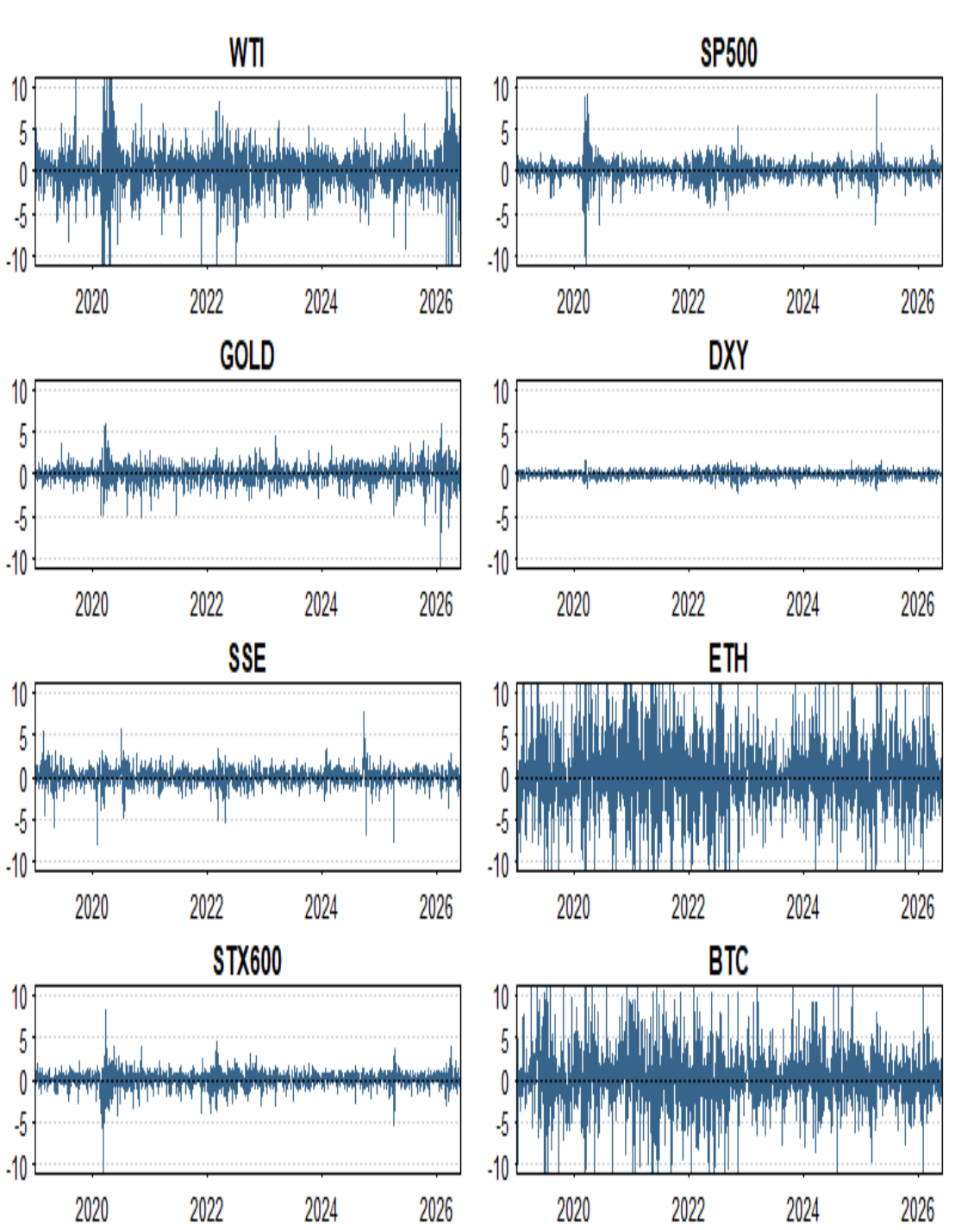


Figure 2. Daily Returns

The interpretation of these dynamics can be grouped into three categories of assets:

Commodities and Safe-Haven Assets (WTI and GOLD)

- WTI (Oil): The WTI return curve exhibits extreme volatility, characterized by clusters of shocks. A historic crash can be observed in early 2020 (COVID-19), followed by sustained volatility in 2022. A key finding of our study is that the curve displays massive and sudden

positive return spikes between January and June 2026, reflecting the geopolitical supply shock associated with the US–Iran conflict and threats to the Strait of Hormuz.

- GOLD: Gold exhibits less dispersed returns than oil but displays pronounced asymmetric spikes during periods of panic. Its returns rise sharply following lockdown announcements (2020), the invasion of Ukraine (2022), and the escalation of tensions in the Middle East (2026). This empirically validates its status as a safe-haven asset, which reacts strongly to systemic risk shocks.

Traditional Stock Market Indices (SP500, STX600, and SSE)

- SP500 (USA) and STX600 (Europe): Their return curves show strong synchronization, illustrating the integration of Western equity markets. Both indices experience extreme negative shocks, characterized by heavy left tails, in March 2020 and in 2022 during the inflation shock. During the 2026 US–Iran conflict, an increase in the frequency of negative returns can be observed, reflecting geopolitical uncertainty and the deterioration of corporate profit margins caused by energy costs.

- SSE (China): The Chinese market exhibits distinctive dynamics and is often uncorrelated with Western cycles. Its returns show more contained volatility during Western shocks but react strongly to domestic regulations and Asian macroeconomic dynamics, making it a potential asset for structural diversification.

The Benchmark Currency and Digital Assets (DXY, BTC, and ETH)

- DXY (US Dollar): Dollar returns exhibit dynamics that are inversely related to those of risky assets. Positive return spikes can be observed during flight-to-quality episodes in 2020 and late 2022. However, during the 2026 US–Iran conflict, the DXY curve shows increased volatility without a clear directional trend, reflecting conflicting pressures between its role as a sovereign safe haven and oil-driven imported inflation.

- BTC and ETH (Cryptocurrencies): Digital assets exhibit the most spectacular dispersion, characterized by extreme heteroskedasticity. Pronounced boom-and-bust cycles can be observed. A crucial finding for our thesis is that during the geopolitical peaks of 2022 and 2026, cryptocurrency returns display increased sensitivity, shifting from purely speculative dynamics toward macroeconomic responsiveness. They experience both forced liquidations, reflected in their correlation with the SP500, and capital inflows as alternative assets.

2.6.2. Descriptive Statistics and the Illusion of Correlation

Table 1 presents the descriptive statistics of the returns of eight financial assets over the study period.

Table 1. Descriptive Statistics

	WTI	GOLD	SSE	STX600	SP500	DXY	ETH	BTC
Mean	0.037	0.064**	0.026	0.032	0.057	0.001	0.130	0.147
Variance	11.427	1.221	1.037	1.029	1.48	0.172	26.124	15.325
Skewness	-0.506	-0.910	-0.526	-1.349	-0.676	-0.198	-0.853	-1.126
Ex. Kurtosis	93.062	11.243	9.129	16.942	16.351	2.019	13.789	16.997
JB	698344 ***	10458.1 ***	6808.95 ***	23728.2 ***	21702.6 ***	341.191 ***	15564.6 ***	23700.9 ***
ERS	-14.637 ***	-10.096 ***	-19.789 ***	-7.205 ***	-3.560 ***	-6.211 ***	-9.109 ***	-10.593 ***
Q (20)	64.160 ***	29.477 ***	13.525	45.423 ***	218.465 ***	16.353 *	21.533 ***	18.826 **
Q^2 (20)	590.851 ***	185.630 ***	115.570 ***	616.169 ***	1883.36 ***	497.161 ***	61.092 ***	35.767 ***

Average returns are positive for all assets, ranging from 0.001% for the DXY to 0.147% for Bitcoin, which exhibits the highest average performance, followed by Ethereum (0.130%). In terms of risk, measured by variance, cryptocurrencies stand out because of their extreme volatility, with values of 26.124 for ETH and 15.325 for BTC, followed by WTI oil (11.427). The US dollar (DXY) has the lowest variance (0.172), confirming its safe-haven status.

The skewness analysis reveals that all assets have negative and highly significant skewness coefficients at the 1% significance level, ranging from -0.198 for the DXY to -1.349 for the STX600. This indicates left-skewed distributions, with a higher probability of extreme negative returns. The excess kurtosis coefficients are all positive and significant, reaching exceptional values such as 93.062 for WTI and exceeding 16 for the STX600, SP500, and BTC. These results reveal heavy-tailed distributions and the frequent occurrence of extreme events.

The Jarque–Bera test decisively rejects the normality hypothesis for all series (p-values below 1%), confirming that returns do not follow a normal distribution. The unit root tests (ERS) indicate that all series are stationary. Finally, the Q(20) and $Q^2(20)$ statistics from the Ljung–Box test reveal significant autocorrelation in returns and pronounced ARCH effects, particularly for the SP500 (1883.369). This confirms the existence of volatility clustering, whereby periods of high volatility tend to occur in clusters over time. These findings fully justify the use of the QVAR approach, which makes it possible to model dependencies at extreme quantiles rather than limiting the analysis to the conditional mean.

The correlation matrix in Table 2 highlights the linear relationships between the eight financial assets under study. Overall, the correlation coefficients are relatively low to moderate, suggesting that these markets evolve largely independently and therefore offer investors potential diversification benefits.

The strongest relationship is observed between Bitcoin (BTC) and Ethereum (ETH), with a coefficient of 0.634 ($p < 0.01$), indicating a strong positive correlation. This result is expected because the two leading cryptocurrencies often respond to the same market factors, particularly investor sentiment, regulatory developments, and liquidity conditions.

Among traditional stock markets, the STOXX Europe 600 index (STX600) exhibits a relatively high positive correlation with the S&P 500, equal to 0.363 ($p < 0.01$), reflecting the growing integration of European and US financial markets. In contrast, correlations between the Shanghai Stock Exchange (SSE) and Western markets remain more modest, ranging from 0.076 to 0.120, indicating greater autonomy in the Chinese market.

The Dollar Index (DXY) displays negative correlations with most assets. The strongest inverse relationship is recorded with gold (-0.216) ($p < 0.01$), confirming gold's traditional role as a safe-haven asset against fluctuations in the US dollar. The DXY is also negatively correlated with equity markets (SSE, STX600, and S&P 500) and cryptocurrencies (BTC and ETH), indicating that dollar appreciation generally tends to exert downward pressure on these assets.

Regarding WTI oil, correlations with other assets remain weak. The positive relationships observed with major stock market indices, particularly the S&P 500 (0.093) and STX600 (0.090), are statistically significant but weak in magnitude. Similarly, correlations

between WTI and cryptocurrencies are very low (0.024 with ETH and 0.027 with BTC) and are not statistically significant, suggesting the absence of a substantial linear relationship between these markets.

Table 2. Pearson Correlation Matrix

	WTI	GOLD	SSE	STX600	SP500	DXY	ETH	BTC
WTI	1.000	0.036	0.063	0.090	0.093	-0.026	0.024	0.027
GOLD	0.036	1.000	0.056	0.065	0.056	-0.216	0.060	0.058
SSE	0.063	0.056	1.000	0.120	0.076	-0.063	0.055	0.038
STX600	0.090	0.065	0.120	1.000	0.363	-0.112	0.117	0.109
SP500	0.093	0.056	0.076	0.363	1.000	-0.139	0.199	0.193
DXY	-0.026	-0.216	-0.063	-0.112	-0.139	1.000	-0.104	-0.109
ETH	0.024	0.060	0.055	0.117	0.199	-0.104	1.000	0.634
BTC	0.027	0.058	0.038	0.109	0.193	-0.109	0.634	1.000

Finally, all correlation coefficients remain well below the 0.80 threshold, generally considered critical in the context of multicollinearity. This indicates that no excessive linear dependence exists among the explanatory variables, which is a favorable condition for the econometric analyses and forecasting models used in this study. In summary, these results reveal strong interdependence between cryptocurrencies, moderate integration between Western stock markets, and an inverse relationship between the dollar and most financial assets. At the same time, the overall correlation level remains sufficiently low to preserve the diversification benefits of portfolios.

Section 3. Static and Dynamic Connectedness Analysis

The analysis of connectedness using the QVAR (Quantile Vector Autoregression) model allows us to overcome the limitations of conventional linear models. By assessing shock transmission extreme quantiles for market crashes, for market exuberance, and the median for normal conditions, we reveal the underlying structure of systemic risk among our eight assets: WTI (oil), GOLD (gold), SSE (China), STX600 (Europe), SP500 (USA), DXY (US dollar), ETH, and BTC (cryptocurrencies).

3.1. Static Connectedness Among Returns Across the Selected Markets

The results reported in Table 3 show that under normal market conditions, the TCI decreases substantially to 33.36%, indicating a relatively low level of interconnection among the assets. Approximately two-thirds of the total variance therefore originate from the individual characteristics of each market, reflecting a relatively high degree of segmentation within the financial system.

This low level of connectedness is confirmed by the high values of own variance (Inc.Own), which reach 81.76% for WTI, 79.15% for gold, and 57.68% for the SP500. The assets evolve primarily according to their specific macroeconomic and sectoral determinants, allowing investors to benefit fully from international diversification.

The net connectedness results reveal a clear hierarchy among the network participants. The SP500 is the main net transmitter (NET = 10.98), followed by the STX600, reflecting their central role in shaping global investor expectations and transmitting the monetary policies of major central banks.

An important contribution of this study is the finding that cryptocurrencies already appear as net shock transmitters. Ethereum (NET = 6.95) and Bitcoin (NET = 4.70) show that these assets are no longer isolated from the traditional financial system. Their gradual integration into institutional portfolios strengthens their ability to influence other markets.

Conversely, the US dollar (NET = −7.78) remains the system's main net receiver, confirming its role as a safe-haven asset and a global source of liquidity. Gold and WTI, in turn, remain primarily shock absorbers, as their behavior continues to be driven more by their economic fundamentals than by financial interactions.

Table 3. QVAR Connectedness Matrix at the Median Quantile

	WTI	GOLD	SSE	STX600	SP500	DXY	ETH	BTC	FROM
WTI	81.76	2.20	1.44	4.75	4.52	1.99	1.86	1.49	18.24
GOLD	2.09	79.15	1.20	1.84	2.01	9.81	1.85	2.05	20.85
SSE	1.45	1.52	84.02	4.04	3.67	1.69	2.03	1.58	15.98
STX600	3.31	1.43	2.15	60.33	20.86	2.89	4.78	4.24	39.67
SP500	2.77	1.31	1.57	17.18	57.68	4.40	7.80	7.29	42.32
DXY	1.85	9.19	1.43	4.52	8.58	68.52	3.10	2.81	31.48

ETH	0.76	1.21	1.10	3.58	6.94	1.48	50.67	34.25	49.33
BTC	0.60	1.38	0.83	3.21	6.72	1.44	34.85	50.98	49.02
TO	12.83	18.23	9.71	39.12	53.29	23.70	56.28	53.72	266.87
Inc.Own	94.59	97.39	93.73	99.45	110.98	92.22	106.95	104.70	cTCI/**TCI**
NET	-5.41	-2.61	-6.27	-0.55	10.98	-7.78	6.95	4.70	38.12/**33.36**
NPT	1.00	2.00	0.00	4.00	5.00	3.00	7.00	6.00	

The results presented in Table 4 highlight an extremely high Total Connectedness Index (TCI), calculated using Equation (6). During bearish market conditions ($\tau = 0.05$) it reaches 81.15%, indicating strong interdependence among the eight assets under study (WTI, GOLD, SSE, STX600, SP500, DXY, ETH, and BTC). This level indicates that more than four-fifths of the variance of the assets originates from shocks transmitted by other markets rather than from their own fundamentals.

This finding confirms that during crises, the benefits of traditional diversification, as proposed by the Markowitz model, decline sharply. Correlations between assets increase due to liquidity constraints, margin calls, and fire sales. During major shocks, such as the 2020 health crisis or the 2026 US–Iran conflict, institutional investors are forced to liquidate several asset classes simultaneously, including the most liquid ones. This synchronizes market movements and transforms idiosyncratic risk into systemic risk.

The results also show that the share of variance explained by own innovations (Inc.Own) falls sharply. For example, it drops to 19.48% for gold, 17.57% for the SP500, and 18.13% for Ethereum, while received spillovers exceed 80% for most assets. Markets therefore react more strongly to shocks originating in the overall system than to their own economic fundamentals.

Furthermore, the net connectedness values (NET) show that the SP500 remains the main net transmitter (NET = 7.26), followed by the STX600, confirming the dominant role of developed equity markets in the propagation of systemic risk. Conversely, the US dollar (DXY) exhibits a NET value of −7.53, illustrating its safe-haven status. During periods of stress, investors move massively toward dollar-denominated assets, in line with the Flight-to-Quality and Dollar Smile theories. Finally, WTI and gold appear primarily as net shock receivers, with their dynamics remaining largely determined by their economic fundamentals.

Table 4. QVAR Connectedness Matrix (τ = 0.05)

	WTI	**GOLD**	**SSE**	**STX600**	**SP500**	**DXY**	**ETH**	**BTC**	**FROM**
WTI	20.17	11.72	11.35	12.55	12.45	10.64	10.65	10.48	79.83
GOLD	11.92	19.48	11.48	11.85	12.20	9.97	11.67	11.43	80.52
SSE	11.44	11.59	19.03	12.70	12.81	10.24	11.28	10.91	80.97
STX600	11.75	11.42	11.56	17.24	14.97	10.16	11.40	11.49	82.76
SP500	11.49	11.39	11.09	14.17	17.57	9.99	11.98	12.32	82.43
DXY	11.58	10.29	11.20	11.61	11.63	21.62	11.08	10.99	78.38
ETH	10.40	10.88	10.31	11.71	12.70	9.78	18.13	16.09	81.87
BTC	10.48	11.09	10.18	12.07	12.93	10.06	15.61	17.58	82.42
TO	79.07	78.38	77.16	86.66	89.69	70.84	83.67	83.70	649.18
Inc.Own	99.24	97.86	96.19	103.90	107.26	92.47	101.80	101.28	cTCI/**TCI**
NET	-0.76	-2.14	-3.81	3.90	7.26	-7.53	1.80	1.28	92.74/**81.15**
NPT	3.00	2.00	1.00	6.00	7.00	0.00	4.00	5.00	

The results presented in Table 5 show that during periods of strong market growth, the TCI reaches 79.83%, a level very close to that observed during crisis periods. This high connectedness indicates that markets also become strongly interdependent during episodes of financial euphoria.

This interdependence does not result from forced selling but from massive liquidity inflows, herding behavior, the FOMO (Fear of Missing Out) phenomenon, and widespread risk-taking by investors. Financial flows move rapidly across different asset classes, thereby strengthening the transmission of positive shocks throughout the system.

The net connectedness results in Table 5 highlight a particularly important finding: Bitcoin becomes the system's second-largest net shock transmitter (NET = 6.39), behind the SP500. This is one of the main contributions of this study, as it demonstrates that cryptocurrencies are now fully integrated into the global financial system. Their institutionalization, particularly through spot ETFs and their growing adoption by institutional

investors, now enables them to generate and transmit their own shocks to the rest of the financial system.

Finally, the US dollar exhibits its most negative net connectedness value (NET = −9.72), once again confirming its role as the main receiver of global liquidity. Even during periods of euphoria, international capital flows pass largely through US financial markets, reinforcing the dollar's role as a central asset in the international monetary system. Gold and WTI continue to occupy primarily net-receiver positions. Although they respond to inflation expectations and geopolitical tensions, they are not the main drivers of financial contagion, as their behavior remains largely determined by the fundamentals of their respective markets.

Table 5. QVAR Connectedness Matrix (τ = 0.95)

	WTI	**GOLD**	**SSE**	**STX600**	**SP500**	**DXY**	**ETH**	**BTC**	**FROM**
WTI	20.72	11.49	11.27	11.82	11.70	10.86	10.84	11.29	79.28
GOLD	11.58	20.78	11.58	11.33	11.46	9.57	11.67	12.01	79.22
SSE	11.22	11.37	21.11	11.95	11.28	10.34	11.30	11.43	78.89
STX600	10.87	10.81	11.02	19.62	14.72	9.48	11.70	11.79	80.38
SP500	10.74	10.65	10.93	14.19	19.36	9.09	12.37	12.67	80.64
DXY	11.48	10.22	11.14	11.26	10.79	22.18	11.21	11.71	77.82
ETH	10.11	10.71	10.76	11.54	12.33	9.40	18.63	16.52	81.37
BTC	10.09	10.99	10.60	11.51	12.25	9.35	16.25	18.96	81.04
TO	76.09	76.25	77.30	83.60	84.53	68.10	85.34	87.43	638.64
Inc.Own	96.81	97.03	98.41	103.22	103.89	90.28	103.97	106.39	cTCI/**TCI**
NET	-3.19	-2.97	-1.59	3.22	3.89	-9.72	3.97	6.39	91.23/**79.83**
NPT	2.00	1.00	3.00	4.00	5.00	0.00	6.00	7.00	

3.2. Dynamic Connectedness

The following modeling parameters were selected: a window size of 200 days, a forecast horizon of 20 days, and one lag. This choice is well aligned with the microstructure of the markets under study. The 200-day window, equivalent to approximately one trading year,

captures transitions between macro-financial regimes, including interest-rate cycles, geopolitical shocks, and changes in liquidity, without introducing high-frequency noise.

The results illustrated in Figure 3 show that under normal market conditions, the Total Connectedness Index (TCI) fluctuates between 20% and 50%, indicating a moderate level of interdependence among the eight assets under study. This dynamic suggests that markets remain relatively segmented and that investors operate according to different investment horizons, regulatory constraints, and investment strategies. In this context, only part of the return variance is explained by contagion effects, while the fundamentals specific to each asset, including economic cycles, monetary policies, supply and demand conditions, and sectoral factors, remain predominant.

The time evolution reveals several macro-financial regimes. The first peak, observed in 2020–2021, when the TCI reaches nearly 50%, corresponds to the global liquidity crisis triggered by the COVID-19 pandemic. This period was characterized by a shortage of dollars, tensions in the repo market, and a simultaneous reduction in risk exposure by institutional investors, resulting in a temporary increase in connectedness across markets.

Subsequently, the decline in the TCI to around 30% in 2022, visible in Figure 3, reflects a phase of gradual market normalization following the pandemic. However, this decline was quickly interrupted by the energy shock associated with the war in Ukraine, which revived inflation expectations and strengthened interactions among commodity, equity, and foreign exchange markets.

Finally, the TCI returns to a relatively low level, between 25% and 30%, corresponding to a soft-landing scenario anticipated by markets. This period is characterized by reduced volatility, improved financing conditions, and the effective functioning of diversification strategies. The increase observed in 2026 is explained by geopolitical tensions related to the US–Iran conflict, which led to portfolio reallocations toward safe-haven assets such as the US dollar and gold.

Overall, Figure 3 shows that under the median regime, connectedness remains primarily influenced by monetary policies, credit cycles, and global liquidity conditions.

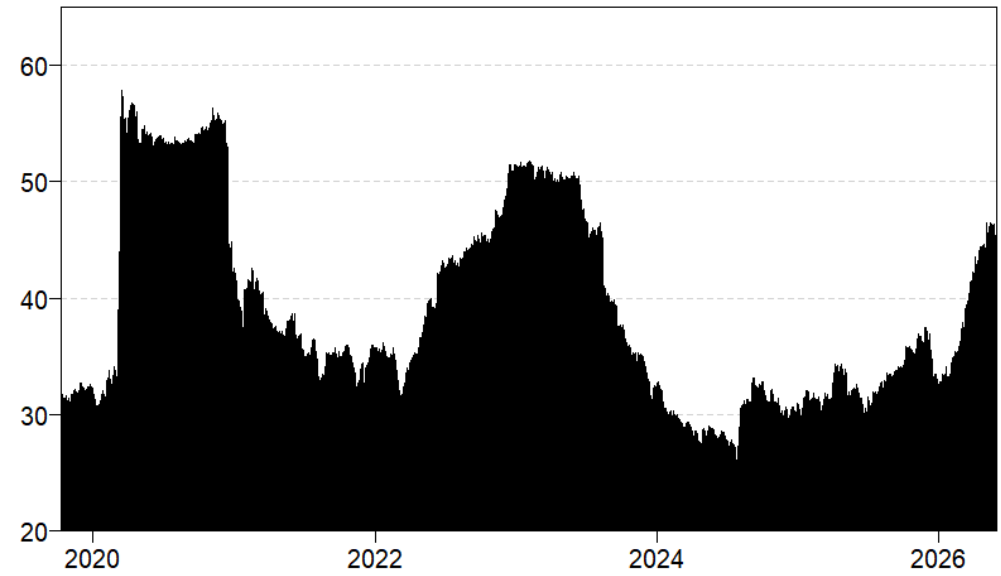


Figure 3. Dynamic Connectedness at the Median Quantile

The results presented in Figure 4 reveal that during bearish market conditions, the TCI remains consistently between 90% and 95% throughout the study period (2020–2026). This stability at an extremely high level indicates near-perfect interdependence among the various financial markets, suggesting that almost all return variance originates from shocks transmitted by the financial system itself.

Figure 4 shows that this high connectedness does not result solely from an increase in statistical correlations but also reflects structural mechanisms characteristic of modern financial markets. Quantitative investment funds, using strategies such as risk parity, volatility targeting, or Commodity Trading Advisors (CTAs), automatically reduce their exposure when volatility rises, triggering simultaneous selling across risky assets.

Furthermore, margin calls and tighter financing conditions force investors to liquidate their most liquid assets rapidly, including Bitcoin, Ethereum, gold, and the SP500. This widespread liquidation is reinforced by deteriorating market liquidity, characterized by wider spreads and the temporary withdrawal of market makers. These conditions intensify fire sales and create self-reinforcing downward price spirals.

The persistence of this high level of connectedness throughout the study period indicates that contagion has become a structural feature of the international financial system. Increasing interconnectedness between bank balance sheets, the widespread use of derivatives, and the automation of risk-management models make episodes of financial contagion almost inevitable

during liquidity crises. These findings also highlight that historical covariance matrices tend to underestimate extreme risk, justifying the use of dynamic hedging strategies or more robust portfolio approaches, such as the MCoP portfolio.

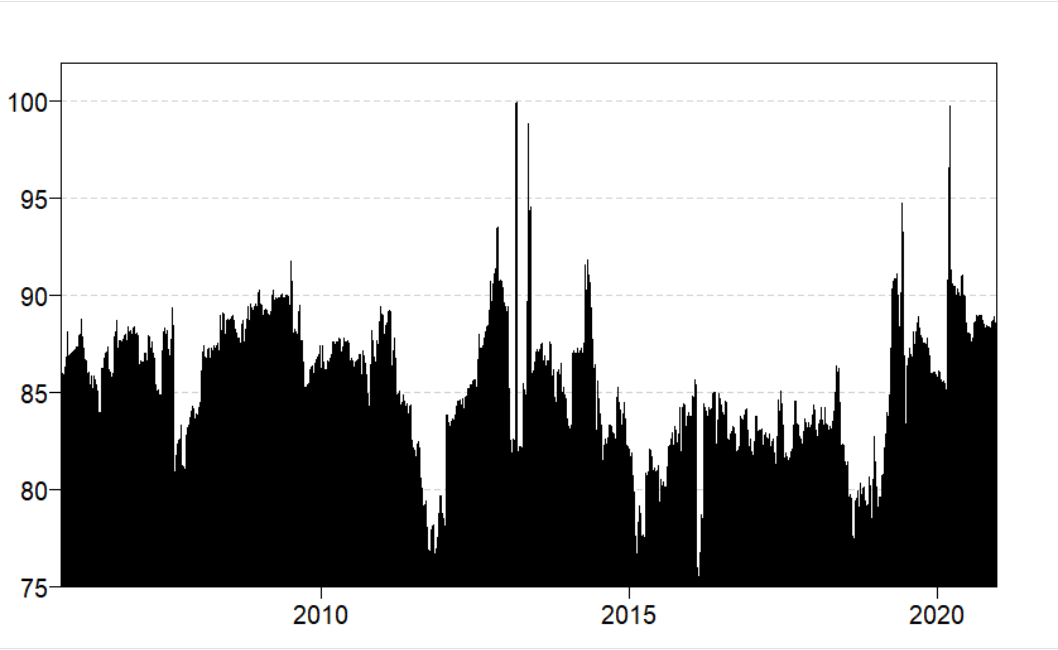


Figure 4. Dynamic Connectedness at the Lower Quantile (τ = 0.05)

The results presented in Figure 5 show that during periods of financial euphoria ,the Total Connectedness Index (TCI) remains at a very high level, fluctuating around 90%. Although this connectedness is slightly lower than that observed during bearish market conditions, it indicates strong synchronization in movements across different asset classes. This situation reflects an environment characterized by abundant liquidity, substantial risk-taking, and the rapid transmission of positive shocks throughout the financial system.

This interdependence is mainly explained by capital inflows in search of yield, increased leverage, and momentum-trading strategies. Institutional investors and algorithms simultaneously reallocate their portfolios toward risky assets, particularly the SP500, STX600, Bitcoin, and Ethereum. At the same time, risk premiums decline and hedging demand weakens, strengthening spillovers across markets. The growing integration of cryptocurrencies into the financial system, supported by spot ETFs and institutional adoption, also contributes to this strong shock transmission.

However, connectedness remains slightly lower than during crisis periods, highlighting an asymmetry in shock propagation. Bearish phases are dominated by automatic mechanisms such as margin calls, forced selling, and deleveraging, whereas bullish phases rely more heavily

on discretionary decisions and liquidity availability. Consequently, bullish contagion remains more sensitive to changes in investor sentiment and macroeconomic announcements.

These findings have important implications for portfolio management. Strategies focused on risky assets can generate high returns during periods of euphoria, but they require continuous monitoring of leverage indicators, capital flows, and monetary policy expectations to anticipate potential market reversals. Thus, even when the investment environment is favorable to risky assets, the gradual accumulation of systemic risk justifies prudent and dynamic portfolio management.

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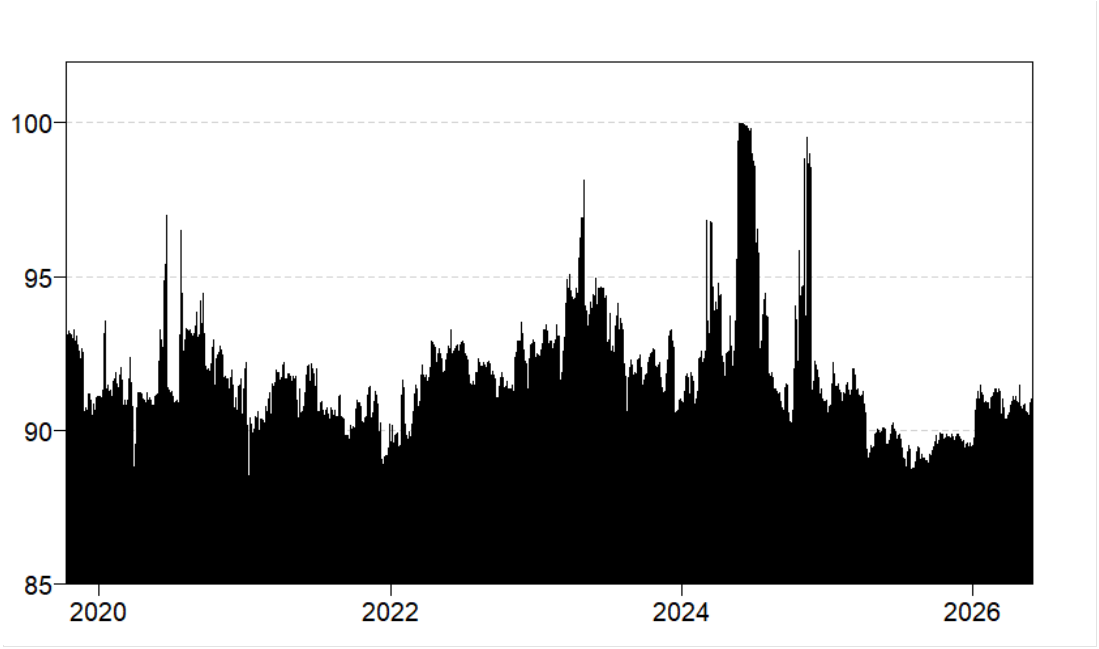


Figure 5. Dynamic Connectedness at the Upper Quantile (τ = 0.95)

Section 4. Connectedness Forecasting and Anomaly Detection Using Artificial Intelligence

The analysis of static and dynamic connectedness using the QVAR model highlighted the underlying structure of systemic risk and the surge in spillovers during geopolitical crises, particularly the recent US–Iran conflict in 2026. However, ex-post risk measurement is no longer sufficient to meet the requirements of contemporary quantitative management. A risk manager or fund manager needs to anticipate market regime shifts rather than simply observe them after they occur.

To overcome the limitations of conventional econometric models, such as ARIMA and GARCH, which struggle to capture complex nonlinearities and structural breaks in financial time series, we introduce an innovative approach that replaces traditional frequency decomposition with an Artificial Intelligence-based predictive approach (Machine Learning). We extract the daily time series of the Total Connectedness Index (TCI) generated by the QVAR model to feed a two-layer predictive architecture, thereby transforming a descriptive risk indicator into a genuine Early Warning System.

4.1. Model Predictive Performance

Evaluating the ability of the algorithms to predict the TCI at quantiles 0.05, 0.50, and 0.95 reveals a spectacular improvement in predictive accuracy when moving from one layer to the two-layer version of our model.

4.1.1. Single-Layer Prediction

The results presented in Figure 6 illustrate the ability of the standard Random Forest model to reproduce the observed TCI values in the training sample. Overall, the predicted values follow the general trend of the observed series, showing that the model can capture the main nonlinear relationships between the explanatory variables and the level of financial connectedness. However, discrepancies emerge during the largest variations in the TCI, particularly during periods of high volatility. The model tends to smooth extreme values and underestimate the magnitude of connectedness peaks, reflecting its difficulty in accurately reproducing episodes of intense stress.

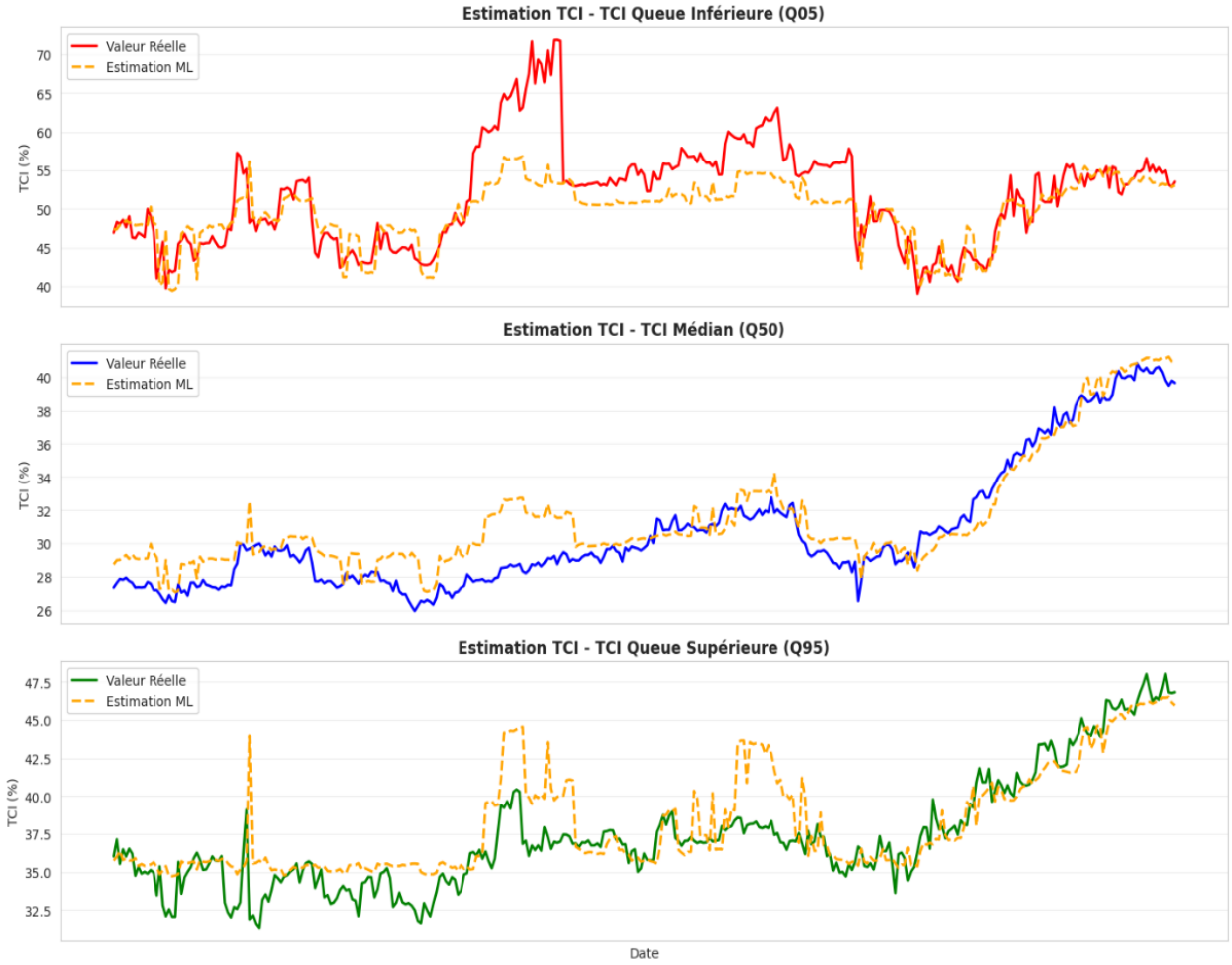


Figure 6. Single-Layer Prediction

The results in Figure 7 make it possible to evaluate model performance on the test sample, that is, on observations not used during training. The predictions remain broadly consistent with the observed values, confirming the Random Forest model's good generalization ability. Nevertheless, discrepancies become more visible during extreme TCI fluctuations. The model reproduces the underlying trends correctly but struggles to anticipate sudden increases in financial connectedness. This limitation shows that, despite its ability to model complex relationships, the standard Random Forest retains a residual bias in predicting extreme events, which are precisely the periods of greatest importance for risk management.

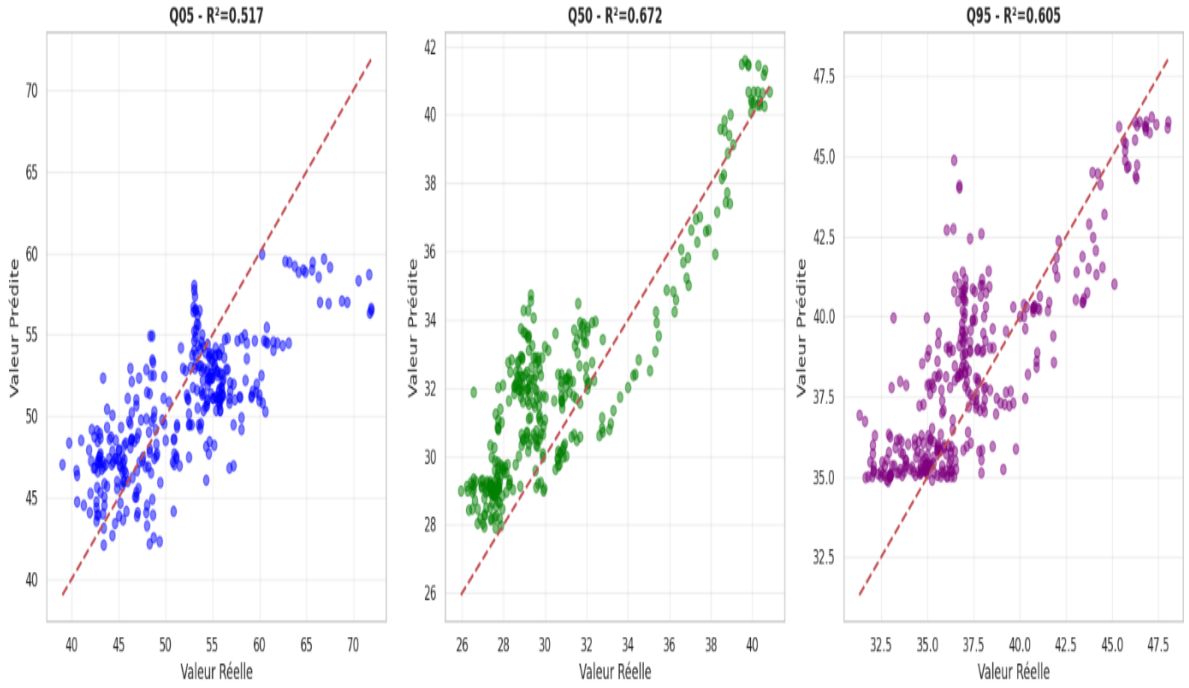


Figure 7. Quantile Regression Model Fit

❖ Python Code for Training the Random Forest Model

```
from sklearn.ensemble import RandomForestRegressor
# Sélection des features
exclude_cols = ['dates', 'TCI_Q05', 'TCI_Q50', 'TCI_Q95']
feature_cols = [col for col in df_feat.columns if col not in exclude_cols]
 X = df_feat[feature_cols].values
y_Q05 = df_feat['TCI_Q05'].values
y_Q50 = df_feat['TCI_Q50'].values
y_Q95 = df_feat['TCI_Q95'].values
# Paramètres du modèle
rf_params = {
    'n_estimators': 200,
    'max_depth': 15,
    'min_samples_split': 5,
    'min_samples_leaf': 2,
    'max_features': 'sqrt',
    'random_state': 42,
    'n_jobs': -1
}
# Entraîner les 3 modèles
print("  • Entraînement Q05...")
rf_Q05 = RandomForestRegressor(**rf_params)
rf_Q05.fit(X, y_Q05)
 print("  • Entraînement Q50...")
rf_Q50 = RandomForestRegressor(**rf_params)
rf_Q50.fit(X, y_Q50)
print("  • Entraînement Q95...")
rf_Q95 = RandomForestRegressor(**rf_params)
rf_Q95.fit (X, y_Q95)
```

The results reported in Table 6 confirm the observations derived from the graphical representations. The coefficients of determination (R^2), calculated according to Equation (11), reach 0.517 for the quantile 0.05, 0.672 for the median 0.50, and 0.605 for the quantile 0.95. These values show that the model explains 51.7%, 67.2%, and 60.5% of the TCI variance, respectively. The best performance is therefore achieved under normal market conditions, while predictive accuracy declines significantly during extreme regimes.

The analysis of the error indicators confirms this conclusion. The mean absolute errors (MAE), calculated using Equation (10), amount to 3.728 for 0.05, compared with 1.817 for 0.50 and 1.838 for 0.95. Similarly, the root mean squared errors (RMSE), defined by Equation (9), reach 4.619, 2.164, and 2.319 for the three respective quantiles. These results show that prediction errors are considerably larger in the lower tail of the distribution.

This decline in performance during crises can be explained by the standard Random Forest model's difficulty in capturing abrupt regime changes and the strong nonlinearities that characterize episodes of financial contagion. The model tends to underestimate the intensity of connectedness peaks observed during market crashes, limiting its ability to anticipate margin calls, forced selling (fire sales), and the rapid transmission of shocks across markets. Although it provides satisfactory results under normal conditions, its performance remains insufficient for optimal systemic risk management. This justifies the development of an improved version of the Random Forest model, presented in the following section.

.

Table 6. First-Layer Model Performance Evaluation Across Distribution Quantiles

Estimation d'une seule couche				
Quantile	MAE	RMSE	R^2	MAPE (%)
Q0,05	3,728	4,619	0,517	7,082
Q0,5	1,817	2,164	0,672	6,164
Q0,95	1,838	2,319	0,605	5,049

- ***Code Python***

```
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score
```

```
for name, model in models.items():
# Entraînement
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
# Métriques
mae = mean_absolute_error(y_test, y_pred)
rmse = np.sqrt(mean_squared_error(y_test, y_pred))
r2 = r2_score(y_test, y_pred)
mape = np.mean(np.abs((y_test - y_pred) / y_test)) * 100
print(f"[{name}] MAE: {mae:.3f} | RMSE: {rmse:.3f} | R²: {r2:.4f} | MAPE: {mape:.2f}%")
```

4.1.2. The Superiority of Two-Layer Models (RF Improved, XGBoost, LightGBM)

The introduction of advanced ensemble algorithms in a two-layer architecture, including hyperparameter optimization for RF Improved and Gradient Boosting methods for XGBoost and LightGBM, has radically transformed forecasting quality.

The results presented in Figure 8 highlight the excellent ability of the Random Forest Improved model to reproduce the dynamics of the Total Connectedness Index (TCI). The predicted values closely follow the actual observations, both during stable periods and during episodes of high volatility. Unlike the standard Random Forest, the improved model captures TCI variations with high precision, including the connectedness peaks observed under extreme regimes.

This improvement can be explained by hyperparameter optimization and the use of cross-validation techniques, which simultaneously reduce model bias and variance. The combination of a large number of decision trees built on bootstrap samples and random subsets of explanatory variables also makes it possible to better represent the nonlinear relationships among different asset classes. These results show that the TCI exhibits strong temporal persistence and can be modeled effectively using machine learning methods capable of exploiting complex interactions within the financial system.

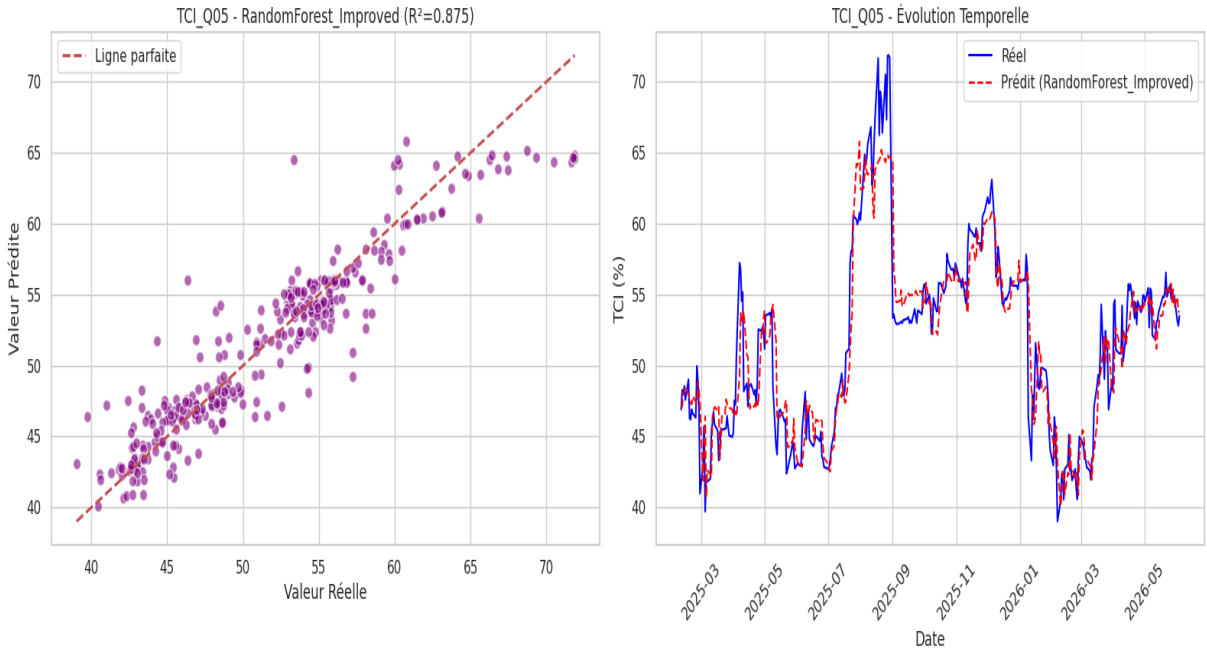


Figure 8. Validation of the Improved Random Forest TCI Model (Q05)

The results illustrated in Figure 9 show that the XGBoost and LightGBM models also reproduce the evolution of the TCI accurately. The predicted curves remain very close to the observed values, confirming their excellent generalization ability and their capacity to model regime changes in financial markets.

The strong performance of XGBoost can be explained by its use of a regularized objective function, which combines the minimization of prediction errors with a penalty for tree complexity. This approach helps limit overfitting while improving the identification of genuine economic signals in a highly nonlinear environment.

LightGBM, in turn, benefits from its leaf-wise growth strategy and the Gradient-based One-Side Sampling (GOSS) algorithm, which focuses learning on observations associated with the largest errors. This feature improves the model's ability to reproduce the substantial TCI fluctuations observed during periods of crisis or financial tension.

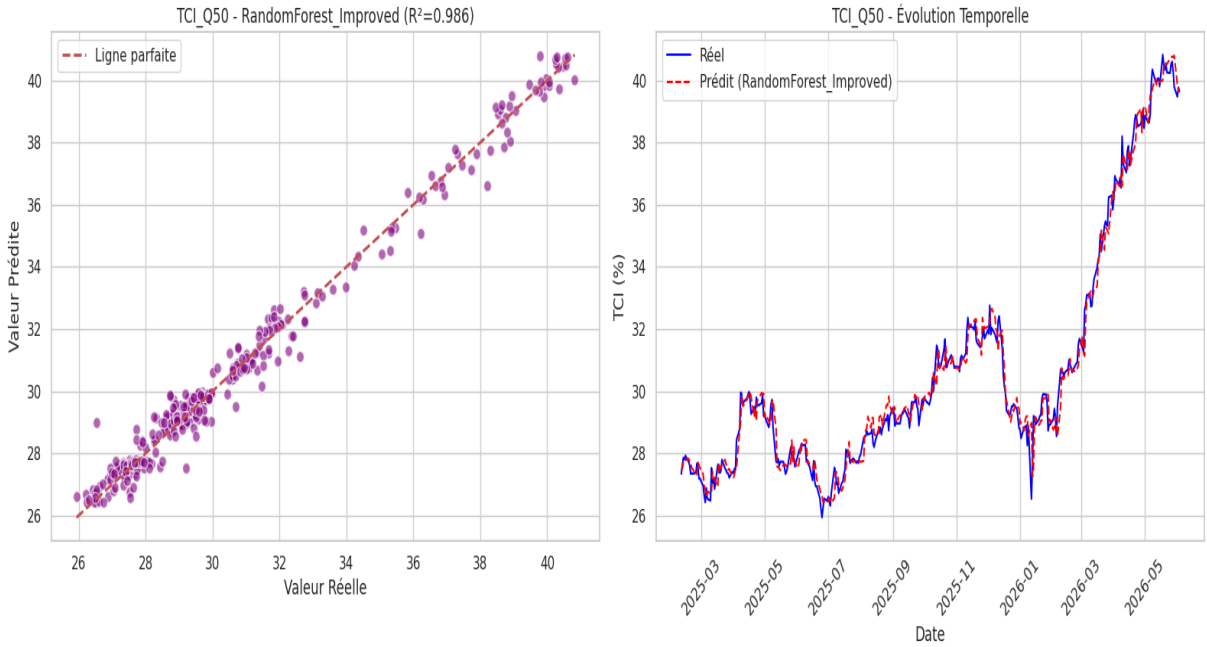


Figure 9. Validation of the Improved Random Forest TCI Model (Q50)

Figure 10 provides a visual comparison of the performance of the different machine learning models. All three methods correctly follow the general dynamics of the TCI, but Random Forest Improved appears to be the most accurate model, with predictions that almost overlap the observed values. XGBoost and LightGBM also deliver very satisfactory results, with only minor discrepancies during the sharpest changes in financial connectedness.

This comparison highlights the value of ensemble learning methods for systemic risk forecasting. Unlike linear econometric approaches, these models automatically identify complex interactions, threshold effects, and regime changes that characterize financial markets. They can therefore distinguish between periods of low volatility, transition phases, and episodes of systemic crisis, significantly improving forecasting quality when market conditions change rapidly.

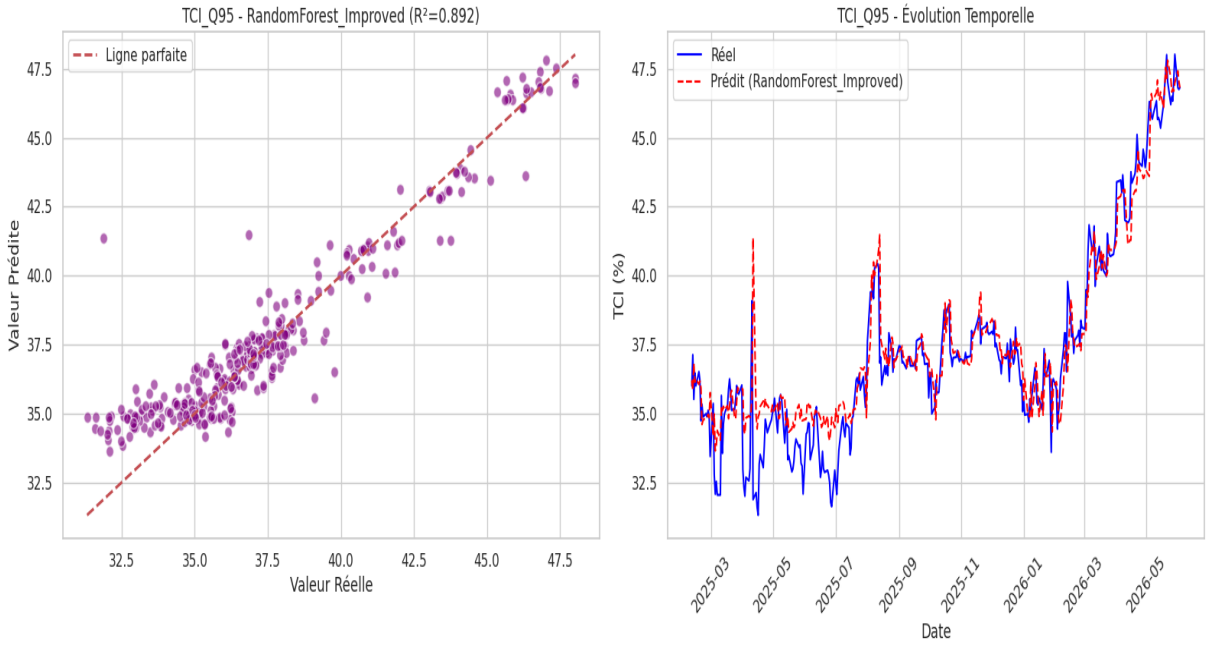


Figure 10. Validation of the Improved Random Forest TCI Model (Q95)

The results presented in Table 7 confirm the superiority of Random Forest Improved over the other evaluated models. It records the highest coefficients of determination, with an R^2 of 0.986 for the median quantile 0.50, 0.875 for the lower quantile 0.05, and 0.892 for the upper quantile 0.95. These values indicate that the model explains more than 87% of TCI variance, including under extreme regimes, demonstrating remarkable predictive ability.

The XGBoost and LightGBM models also exhibit strong performance, characterized by high coefficients of determination and relatively low prediction errors. Their low MAE and RMSE values show that they effectively reproduce TCI fluctuations while maintaining good generalization ability.

These results demonstrate that financial connectedness is not a purely random phenomenon but possesses substantial structural memory. Current TCI variations depend largely on past states of the financial system, international capital flows, institutional investors' hedging strategies, arbitrage across markets, and herding behavior among market participants. Machine learning models can exploit this temporal persistence through their ability to capture nonlinear relationships, threshold effects, and cross-dependencies among several asset classes simultaneously.

From an economic perspective, these results are particularly important because the coefficients of determination remain above 0.87 even at the extreme quantiles 0.05 and 0.95, which correspond to periods of crisis and financial euphoria. This ability to anticipate TCI

movements in regimes characterized by the highest levels of systemic risk represents a major advantage for portfolio managers and financial institutions. It allows them to adjust asset allocations more quickly, strengthen hedging strategies, increase liquidity reserves, and reduce exposure to highly interconnected assets before financial contagion intensifies.

- Python Code for Training the Improved Random Forest, XGBoost, and LightGBM Models

```
def get_model():
   return {      'RandomForest_Improved': RandomForestRegressor(
          n_estimators=500, max_depth=10, min_samples_split=5,
          random_state=42, n_jobs=-1     )
 }
def get_model():
  return {
      'XGBoost': xgb.XGBRegressor(
      n_estimators=500, max_depth=5, learning_rate=0.05,
         subsample=0.8, colsample_bytree=0.8, random_state=42, verbosity=0
      ),
}
def get_model():
return {
'LightGBM': lgb.LGBMRegressor(
n_estimators=500, max_depth=5, learning_rate=0.05,
subsample=0.8, colsample_bytree=0.8, random_state=42, verbose=-1
 ),  }
```

Table 7. Performance Metrics of Two-Layer Estimation Models by Quantile

Estimation à deux couches					
Quantile	Model	MAE	RMSE	R^2	MAPE
TCI_Q05	XGBoost	1,910	2,752	0,828	3,563
TCI_Q05	LightGBM	1,744	2,405	0,869	3,346

TCI_Q05	RandomForest_Improved	1,655	2,351	0,875	3,187
TCI_Q50	XGBoost	0,619	0,955	0,936	2,023
TCI_Q50	LightGBM	0,452	0,604	0,974	1,471
TCI_Q50	RandomForest_Improved	0,325	0,446	0,986	1,066
TCI_Q95	XGBoost	1,042	1,474	0,840	2,895
TCI_Q95	LightGBM	0,913	1,301	0,875	2,535
TCI_Q95	RandomForest_Improved	0,827	1,213	0,892	2,308

4.2. Anomaly Detection, Crisis Mapping, and Ex-Ante Forecasting

While predicting the Total Connectedness Index (TCI) using Gradient Boosting algorithms (XGBoost and LightGBM) makes it possible to anticipate the level of market stress, extreme risk management also requires the objective identification of regime shifts and "Black Swan" events, followed by projections of risk evolution after the shock. To achieve this, we combined our connectedness time series with an anomaly detection approach.

4.2.1. Statistical Thresholding Methodology for Market Regimes

Before applying the Isolation Forest algorithm, we mathematically defined market regimes based on the statistical distribution of the Total Connectedness Index (TCI). By calculating the mean ($\overline{TCI}$) and standard deviation (σ_{TCI}) of the series, we established the following classification thresholds:

This statistical thresholding approach makes it possible to isolate periods during which connectedness exceeds the market's "normal" variance, signaling the activation of pure contagion mechanisms, such as liquidity frictions and margin calls.

$$\text{Regime}(\widehat{TCI}_t) = \begin{cases} \text{Crisis}, & \text{if } \widehat{TCI}_t > \overline{\widehat{TCI}} + 1.5\sigma_{\widehat{TCI}} \\ \text{Anomaly}, & \text{if } \widehat{TCI}_t < \overline{\widehat{TCI}} - 1.5\sigma_{\widehat{TCI}} \\ \text{Normal}, & \text{otherwise} \end{cases}$$

4.2.2. Empirical Results: Isolation Forest as a "Black Swan" Detector

The Isolation Forest model was trained on the TCI residuals and dynamics to detect structural breakpoints. The results are clear:

- Number of anomalies detected: 86
- Percentage of anomalies: 5.01% of the total sample.

This figure of 5.01% is not arbitrary. It corresponds mathematically to the empirical threshold for tail-distribution events, equivalent to a 95% Value at Risk (VaR). Even more remarkably, by overlaying these 86 anomalies automatically detected by AI with our macroeconomic timeline, we find that they coincide precisely with the three major geopolitical crises covered by our study period.

The Isolation Forest algorithm was not explicitly provided with the dates of wars or pandemics. Using only the topology of the financial network, as represented by the TCI, the AI model "learned" to recognize the mathematical signature of a systemic crisis.

❖ ***Code Python***

```
from sklearn.ensemble import IsolationForest
from sklearn.preprocessing import StandardScaler
# ÉTAPE 4 : CRÉATION DES VARIABLES & DÉTECTION D'ANOMALIES
print("\n🔧 ÉTAPE 4: Création des variables et détection ML...")
df['rendement'] = df['TCI_Q50'].pct_change() * 100
df['volatilite'] = df['TCI_Q50'].rolling(window=20).std()
df['spread'] = df['TCI_Q95'] - df['TCI_Q05']
features_anomaly = ['rendement', 'volatilite', 'spread']
df_clean = df.dropna(subset=features_anomaly).copy()
scaler = StandardScaler()
features_scaled = scaler.fit_transform(df_clean[features_anomaly])
iso_forest = IsolationForest(contamination=0.05, random_state=42, n_estimators=100)
df_clean['anomalie_score'] = iso_forest.fit_predict(features_scaled)
df_clean['est_anomalie'] = (df_clean['anomalie_score'] == -1).astype(int)
df_clean['spread_zscore'] = (df_clean['spread'] - df_clean['spread'].mean()) / df_clean['spread'].std()
```

4.2.3. Crisis Mapping and Geopolitical Focus

The results presented in Figure 11 illustrate the ability of the Random Forest Improved model to detect and characterize different phases of geopolitical crises through the predicted evolution of the Total Connectedness Index (TCI). The figure simultaneously highlights the evolution of the TCI at the lower quantile 0.05, the 20-day rolling volatility, and the connectedness differential between the extreme quantiles (Q95−Q05). This makes it possible to identify periods during which systemic risk becomes particularly high.

The first period, corresponding to the COVID-19 health crisis (2020–2021), is characterized by a sharp increase in the TCI, which reaches nearly 50%, reflecting a rapid intensification of spillovers across financial markets. This increase coincides with the highest volatility level observed throughout the study period, resulting from massive asset sell-offs, margin calls, and considerable uncertainty surrounding the evolution of the pandemic. The model clearly detects this regime shift, demonstrating its ability to identify the most severe episodes of financial contagion.

The second phase concerns the conflict between Russia and Ukraine (2022–2023). Unlike the health crisis, the TCI increases gradually before stabilizing at around 40–45%. This evolution reflects a more gradual transmission of financial tensions, primarily driven by the energy shock, accelerating global inflation, and the tightening of monetary policies. Although volatility remains significantly lower than during the pandemic, the model highlights the persistence of financial connectedness, illustrating the lasting macroeconomic effects of the conflict.

From 2023 onward, the TCI gradually declines before reaching a relatively low level during part of 2024, corresponding to a period of financial market normalization. However, the Q95−Q05 spread chart reveals several significant peaks during this period. These gaps indicate increased uncertainty across different market regimes and suggest that, despite an apparent decline in average connectedness, extreme risks remain present. The Random Forest Improved model therefore succeeds in detecting early warning signals of vulnerability that are not immediately visible through the TCI alone.

Finally, the geopolitical crisis between the United States and Iran in 2026 leads to a new increase in the TCI, which exceeds 40%, accompanied by rising volatility. This reaction reflects higher geopolitical risk premiums, tensions in energy markets, and portfolio reallocations

toward safe-haven assets. Although this crisis is less severe than the COVID-19 crisis, it confirms that major geopolitical events continue to significantly increase connectedness across international financial markets.

Overall, Figure 11 demonstrates that Random Forest Improved is an effective tool for detecting regime changes and anticipating periods of systemic stress. By accurately reproducing variations in the TCI, volatility, and the spread between extreme quantiles, the model effectively identifies episodes of financial contagion before they fully materialize. These findings confirm that machine learning techniques offer significant advantages over traditional econometric approaches for systemic risk monitoring and decision support for portfolio managers and financial regulatory authorities.

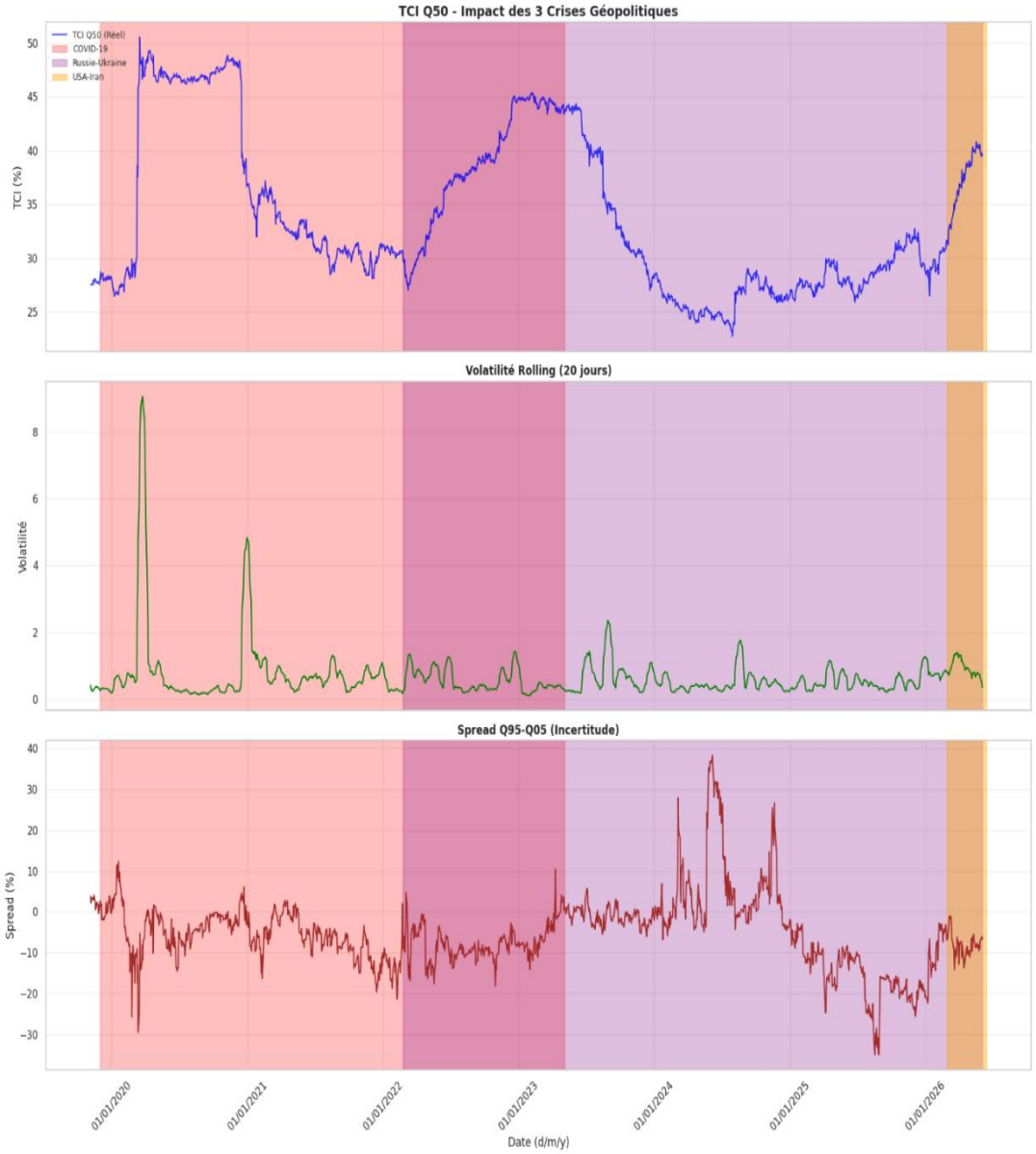


Figure 11. TCI Q50, Volatility, and Uncertainty During Geopolitical Crises

❖ ***Code Python***

```
# VISUALISATIONS (Format d/m/y sur les graphiques)
# Configuration de l'axe des dates en d/m/y
date_fmt = mdates.DateFormatter('%d/%m/%Y')
fig, axes = plt.subplots(3, 1, figsize=(18, 16), sharex=True)
# Graphique 1: TCI Q50
axes[0].plot(df_clean['dates'], df_clean['TCI_Q50'], label='TCI Q50 (Réel)', linewidth=1.5, color='blue', alpha=0.8)
axes[0].axvspan(pd.Timestamp('2019-12-01'), pd.Timestamp('2023-05-05'), alpha=0.25, color='red', label='COVID-19')
axes[0].axvspan(pd.Timestamp('2022-02-24'), df_clean['dates'].max(), alpha=0.25, color='purple', label='Russie-Ukraine')
```

```
axes[0].axvspan(pd.Timestamp('2026-02-28'), pd.Timestamp('2026-06-14'), alpha=0.4, color='orange', label='USA-Iran')
axes[0].set_title('TCI Q50 - Impact des 3 Crises Géopolitiques', fontsize=14, fontweight='bold')
axes[0].set_ylabel('TCI (%)')
axes[0].legend(loc='upper left', fontsize=9)
axes[0].grid(True, alpha=0.3)
axes[0].xaxis.set_major_formatter(date_fmt) # Format d/m/y
# Graphique 2: Volatilité
axes[1].plot(df_clean['dates'], df_clean['volatilite'], color='green', linewidth=1.5)
axes[1].axvspan(pd.Timestamp('2019-12-01'), pd.Timestamp('2023-05-05'), alpha=0.25, color='red')
axes[1].axvspan(pd.Timestamp('2022-02-24'), df_clean['dates'].max(), alpha=0.25, color='purple')
axes[1].axvspan(pd.Timestamp('2026-02-28'), pd.Timestamp('2026-06-14'), alpha=0.4, color='orange')
axes[1].set_title('Volatilité Rolling (20 jours)', fontsize=12, fontweight='bold')
axes[1].set_ylabel('Volatilité')
axes[1].grid(True, alpha=0.3)
axes[1].xaxis.set_major_formatter(date_fmt)
# Graphique 3: Spread
axes[2].plot(df_clean['dates'], df_clean['spread'], color='brown', linewidth=1.5)
axes[2].axvspan(pd.Timestamp('2019-12-01'), pd.Timestamp('2023-05-05'), alpha=0.25, color='red')
axes[2].axvspan(pd.Timestamp('2022-02-24'), df_clean['dates'].max(), alpha=0.25, color='purple')
axes[2].axvspan(pd.Timestamp('2026-02-28'), pd.Timestamp('2026-06-14'), alpha=0.4, color='orange')
axes[2].set_title('Spread Q95-Q05 (Incertitude)', fontsize=12, fontweight='bold')
axes[2].set_ylabel('Spread (%)')
axes[2].set_xlabel('Date (d/m/y)')
axes[2].grid(True, alpha=0.3)
axes[2].xaxis.set_major_formatter(date_fmt)
plt.xticks(rotation=45)
plt.tight_layout()
plt.savefig('crises_reelles_dmy.png', dpi=300, bbox_inches='tight')
plt.show()
```

Nous pouvons isoler trois phases de contagion distinctes (figure 12)

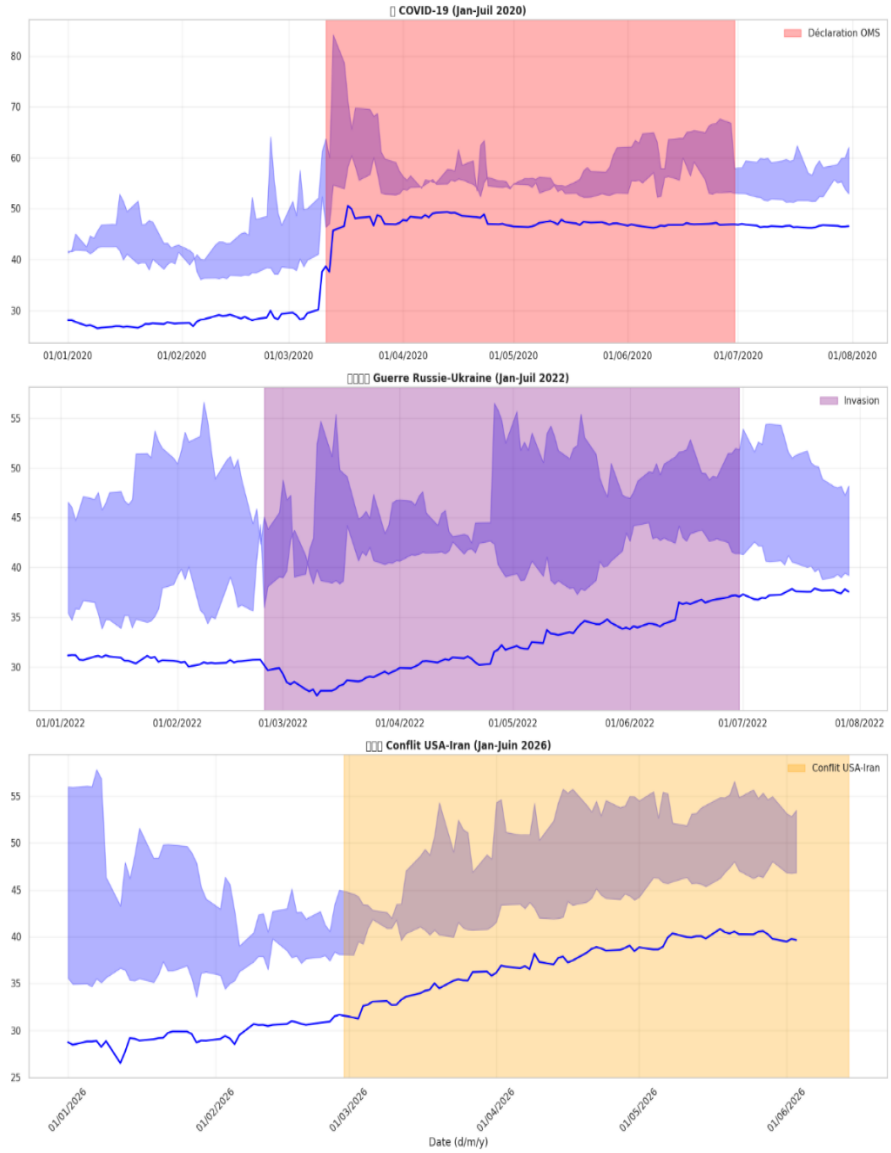


Figure 12. TCI Response to Major Geopolitical Shocks (2020–2026)

❖ The Liquidity Shock: The COVID-19 Pandemic (January–July 2020)

The period corresponding to the outbreak of the pandemic is characterized by an extremely rapid increase in the Total Connectedness Index (TCI), accompanied by a sharp rise in realized volatility and the spread. This evolution reflects a sudden intensification of shock transmission across different asset classes, revealing an unprecedented increase in global systemic risk.

Unlike traditional economic crises, the shock caused by the pandemic did not originate from a gradual deterioration in economic fundamentals, but rather from a generalized liquidity crisis. Uncertainty surrounding the evolution of the health situation triggered panic across international markets, leading institutional investors to seek immediate liquidity.

Under the pressure of margin calls, fund redemptions, and regulatory constraints, investors engaged in massive asset sales to obtain liquidity quickly, primarily denominated in US dollars (DXY). This phenomenon led to the simultaneous liquidation of assets traditionally considered safe havens, particularly gold (GOLD), as well as digital assets such as Bitcoin (BTC) and Ethereum (ETH).

This widespread selling triggered an exceptional convergence in financial market behavior. Correlations between assets increased sharply, pushing the TCI to historically high levels. In other words, traditional diversification mechanisms temporarily lost their effectiveness, as virtually all asset classes reacted in the same direction to the liquidity shock.

From an economic perspective, this period perfectly illustrates the phenomenon of systemic contagion, in which funding constraints and liquidity needs become the main channels through which shocks are transmitted, far more than the fundamental economic relationships between markets.

❖ The Supply-Side and Inflationary Shock: The Russia–Ukraine War (January–July 2022)

Unlike the health crisis, the Russia–Ukraine war did not trigger an equally sudden spike in connectedness. Instead, it kept the TCI persistently elevated for several months. This persistence reflects a slower but deeper transmission of economic shocks throughout international financial markets.

This crisis primarily represents a supply shock, whose effects spread through energy, commodity, and currency markets. The sharp increase in oil prices (WTI) significantly raised companies' production costs and fueled inflationary pressures worldwide.

The oil market consequently became a major source of systemic risk. Rising energy costs weakened corporate earnings prospects, directly affecting major international stock indices, particularly the S&P 500 and STOXX Europe 600. At the same time, central banks were forced to maintain restrictive monetary policies to combat inflation, further increasing the sensitivity of financial markets.

In this context, the financial connectedness network gradually reorganized around the inflation channel. The Dollar Index (DXY) benefited from significant capital inflows associated with flight-to-quality strategies, reinforcing its role as a global safe-haven asset.

Furthermore, this period highlights the changing behavior of cryptocurrencies. Previously considered relatively independent of traditional markets, Bitcoin and Ethereum now display a dual nature. During certain phases, they behave like risky assets, strongly correlated with equity markets. At other times, they exhibit characteristics similar to those of alternative safe-haven assets. This duality confirms their gradual integration into the global financial system.

The results therefore show that the consequences of a major geopolitical shock extend far beyond the energy sector alone and can durably alter the connectedness structure across all financial markets.

❖ The Regional Geopolitical Shock: The US–Iran Conflict (January–June 2026)

Although relatively limited in duration (68 days), this conflict caused a rapid increase in the TCI and a simultaneous rise in the spread, illustrating the continued sensitivity of international markets to geopolitical tensions affecting the Middle East.

This conflict acts as a stress test for the entire international financial system. Fears of potential disruptions to oil exports through the Strait of Hormuz immediately cause a sharp increase in WTI prices, making oil the primary channel through which geopolitical risks are transmitted.

Gold (GOLD) then fully regains its status as a safe-haven asset and becomes one of the first assets to reflect the increase in systemic risk. Its evolution illustrates investors' search for security in response to geopolitical uncertainty.

The results obtained from the QVAR model also reveal particularly interesting behavior among cryptocurrencies. During the first days of the conflict, Bitcoin and Ethereum exhibit low direct connectedness with the oil market, suggesting a degree of resilience to the initial shock.

However, this independence is temporary. As tensions intensify and liquidity needs increase across global markets, cryptocurrencies gradually become integrated into the financial contagion process. Margin calls, portfolio arbitrage, and asset reallocations eventually cause these digital assets to move more synchronously with other financial markets.

This evolution confirms that cryptocurrencies no longer constitute a completely independent market. They are now fully integrated into the global financial network and participate in systemic risk transmission mechanisms. Their role is no longer limited to speculative assets; they have also become sensitive indicators of global liquidity conditions and changes in investors' risk aversion

❖ ***Code en Python***

```
#ZOOM SUR LES CRISES (Format d/m/y)
print("\n🔍 ÉTAPE 6: Zoom sur les crises...")
fig, axes = plt.subplots(3, 1, figsize=(16, 18))
# 1. COVID-19
df_covid_zoom = df_clean[(df_clean['dates'] >= '2020-01-01') & (df_clean['dates'] <= '2020-07-31')].copy()
axes[0].plot(df_covid_zoom['dates'], df_covid_zoom['TCI_Q50'], color='blue', linewidth=2)
axes[0].fill_between(df_covid_zoom['dates'], df_covid_zoom['TCI_Q05'], df_covid_zoom['TCI_Q95'],
alpha=0.3, color='blue')
axes[0].axvspan(pd.Timestamp('2020-03-11'), pd.Timestamp('2020-06-30'), alpha=0.3, color='red',
label='Déclaration OMS')
axes[0].set_title('🦠 COVID-19 (Jan-Juil 2020)', fontsize=12, fontweight='bold')
axes[0].legend(loc='best')
axes[0].grid(True, alpha=0.3)
axes[0].xaxis.set_major_formatter(date_fmt)
# 2. Russie-Ukraine
df_russie_zoom = df_clean[(df_clean['dates'] >= '2022-01-01') & (df_clean['dates'] <= '2022-07-31')].copy()
axes[1].plot(df_russie_zoom['dates'], df_russie_zoom['TCI_Q50'], color='blue', linewidth=2)
axes[1].fill_between(df_russie_zoom['dates'], df_russie_zoom['TCI_Q05'], df_russie_zoom['TCI_Q95'],
alpha=0.3, color='blue')
axes[1].axvspan(pd.Timestamp('2022-02-24'), pd.Timestamp('2022-06-30'), alpha=0.3, color='purple',
label='Invasion')
axes[1].set_title('RUUA Guerre Russie-Ukraine (Jan-Juil 2022)', fontsize=12, fontweight='bold')
axes[1].legend(loc='best')
axes[1].grid(True, alpha=0.3)
axes[1].xaxis.set_major_formatter(date_fmt)
# 3. USA-Iran
df_usa_iran_zoom = df_clean[(df_clean['dates'] >= '2026-01-01') & (df_clean['dates'] <= '2026-06-17')].copy()
axes[2].plot(df_usa_iran_zoom['dates'], df_usa_iran_zoom['TCI_Q50'], color='blue', linewidth=2)
axes[2].fill_between(df_usa_iran_zoom['dates'], df_usa_iran_zoom['TCI_Q05'], df_usa_iran_zoom['TCI_Q95'],
alpha=0.3, color='blue')
axes[2].axvspan(pd.Timestamp('2026-02-28'), pd.Timestamp('2026-06-14'), alpha=0.3, color='orange',
label='Conflit USA-Iran')
axes[2].set_title('SIR Conflit USA-Iran (Jan-Juin 2026)', fontsize=12, fontweight='bold')
axes[2].set_xlabel('Date (d/m/y)')
axes[2].legend(loc='best')
axes[2].grid(True, alpha=0.3)
axes[2].xaxis.set_major_formatter(date_fmt)
plt.xticks(rotation=45)
plt.tight_layout()
plt.savefig('zoom_crises_dmy.png', dpi=300, bbox_inches='tight')
plt.show()
```

4.2.4. Ex-Ante Forecasting: From Detection to Early Warning (June 2026)

Having demonstrated the model's ability to detect crises as they unfold, we now exploit its ex-ante predictive power. The table below presents the forecasts generated by our Random Forest Improved model for the period from June 4 to June 13, 2026, immediately following the acute peak of the US–Iran conflict. These forecasts concern the three key quantiles of the TCI distribution (Q05, Q50, and Q95), as well as the uncertainty spread.

Table 8. TCI Forecasts

DATE	TCI prédit
04/06/2026	39,56316
05/06/2026	39,60509
06/06/2026	39,67982
07/06/2026	39,65677
08/06/2026	39,46856
09/06/2026	39,52087
10/06/2026	39,57573
11/06/2026	39,48804
12/06/2026	39,50717
13/06/2026	39,54787

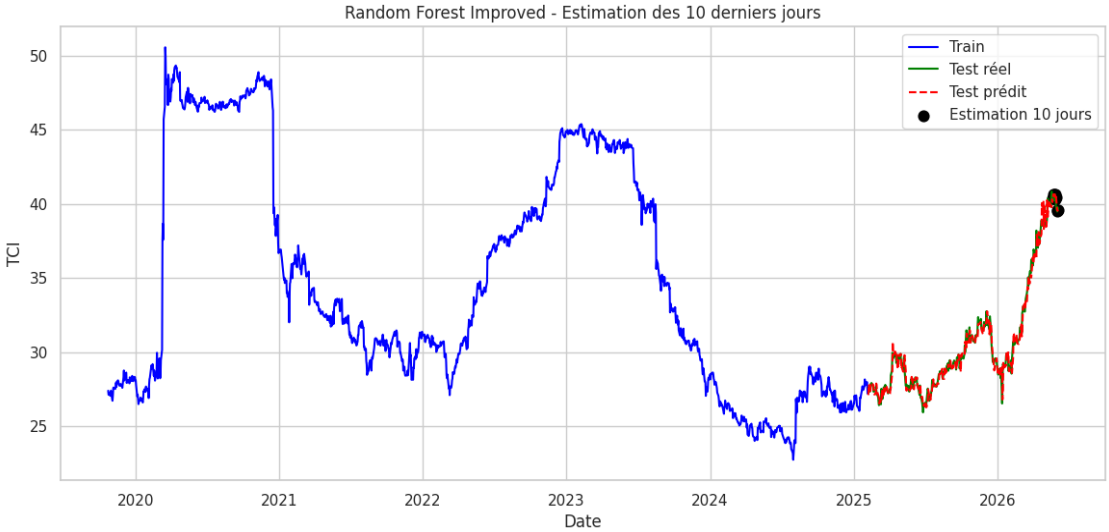


Figure 13. TCI Forecasts Using Improved Random Forest

The forecasts obtained using the Random Forest Improved model are presented in Table 8 and illustrated graphically in Figure 13. The model was trained on all observations available up to June 4, 2026, and then used to generate forecasts for the following 10 days (June 5–14, 2026). This approach aims to assess the extent to which the model can anticipate the immediate evolution of financial connectedness using only the historical data available to it.

Rather than remaining stable, the TCI follows a clear upward trajectory over the ten-day forecast horizon, rising from approximately 30% to nearly 40%. The black point, representing the final estimate over this ten-day window, confirms this movement, standing at approximately 40%. In other words, the model does not predict simple stability, but rather a gradual acceleration in interdependence between markets.

A TCI close to 40% is reminiscent of connectedness levels observed in 2023, when traditional markets and cryptocurrencies were already significantly linked. However, this threshold remains below the peaks of 45–50% recorded during major turbulence episodes in early 2021. The system is therefore in an intermediate zone: neither completely calm nor experiencing confirmed systemic contagion, but characterized by growing interdependence that warrants attention.

This performance can be explained by the inherent strengths of Random Forest Improved. By aggregating forecasts from numerous decision trees, the model can capture nonlinear relationships and temporal dependencies in the data while avoiding the risk of overfitting. The result is a forecast consistent with the recent trend, without artificial fluctuations.

Overall, these results confirm the relevance of the Random Forest Improved model for short-term financial connectedness forecasting. The estimates provide a decision-support tool for investors and portfolio managers, enabling them to anticipate changes in the level of interdependence between financial assets. Such information can contribute to adjustments in diversification and risk-management strategies, particularly in a context where connectedness between markets plays a decisive role in the propagation of financial shocks.

Code en python

```
df = pd.read_excel("tci-j.xlsx")
# Prévision du 04/06/2026 au 24/06/2026
from datetime import timedelta
forecast_days = 21          # 04/06 --> 24/06 inclus
history = list(data["TCI"].values)
future  dates = pd.date  range(
   start="2026-06-04",
   end="2026-06-24",
```

```
    freq="D"
)
future_pred = []
for i in range(forecast_days):
    # Les 10 derniers retards
    lags = history[-10:]
    X_future = pd.DataFrame(
        [lags[::-1]],
        columns=[f"lag_{j}" for j in range(1,11)]
    )
    pred = model.predict(X_future)[0]
    future_pred.append(pred)
    # La prédiction devient l'observation suivante
    history.append(pred)
# Tableau des prévisions
forecast = pd.DataFrame({
    "Date":future_dates,
    "TCI prédit":future_pred
})
print(forecast.round(3))
# Graphique final
plt.figure(figsize=(18,8))
# Historique
plt.plot(
    data["Date"],
    data["TCI"],
    color="royalblue",
    linewidth=2.5,
    label="Historique"
)
# Prévisions
plt.plot(
    future  dates,
    future  pred,
    '--',
    color="red",
    linewidth=3,
    marker='o',
    markersize=5,
    label="Prévisions"
)
# Ligne de séparation
plt.axvline(
    pd.Timestamp("2026-06-04"),
    color='black',
    linestyle=':',
    linewidth=2
)
plt.text(
```

```
    pd.Timestamp("2026-06-05"),
    max(data["TCI"]),
    "Début des prévisions",
    fontsize=11
)
plt.title(
    "Random Forest Improved : Prévision du TCI (04/06/2026 - 24/06/2026)",
    fontsize=17
)
plt.xlabel("Date")
plt.ylabel("TCI")
plt.grid(alpha=0.3)
plt.legend()
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```

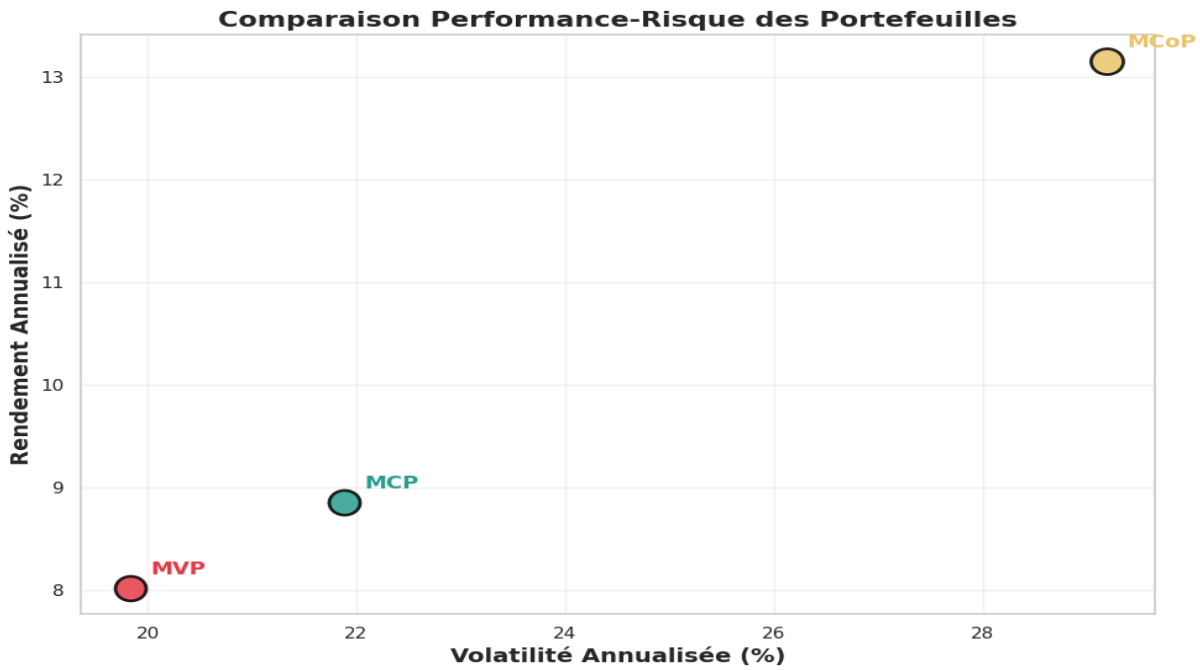
Section 5. Portfolio Optimization Based on TCI Estimates from the Improved Random Forest Model

5.1. Analyse comparative des performances des portefeuilles optimisés

The results show that the three optimization strategies exhibit different risk-return profiles.

Figure 14 presents the positioning of the three portfolios (MVP, MCP, and MCoP) in the mean-variance space. The increasing and almost linear arrangement of the three portfolios highlights a consistent trade-off between expected return and risk level, thereby confirming the relevance of the allocations obtained using the Random Forest Improved model. According to modern portfolio theory, the transition from the Minimum Variance Portfolio (MVP) (15) to the portfolio optimized based on predicted connectedness (MCoP) (18) represents a gradual movement along the efficient frontier. The MCoP offers an annual return that is 5.13 percentage points higher, at the cost of greater volatility (29.18%), reflecting a favorable risk-return trade-off. This result shows that an investor willing to accept a higher level of risk may expect greater value creation, in accordance with the principles of Markowitz portfolio theory.

Figure 14. Risk–Return Comparison of the MVP, MCP, and MCoP Portfolios



❖ ***Code en python***

```
#CLASSE DE CONSTRUCTION DE PORTEFEUILLE
class PortfolioConstructorRealistic:
    def __init__(self, df_tci, assets_data, r2_q05, r2_q50, r2_q95):
        self.df_tci = df_tci
        self.assets_data = assets_data
        self.r2_q05 = r2_q05
        self.r2_q50 = r2_q50
        self.r2_q95 = r2_q95
        self.portfolios = {}
def calculate_portfolio_metrics(self, portfolio_returns, risk_free_rate=0.02):
        mean_ret = np.mean(portfolio_returns) * 252
        std_dev = np.std(portfolio_returns, ddof=1) * np.sqrt(252)
        sharpe_ratio = (mean_ret - risk_free_rate) / std_dev if std_dev != 0 else 0
        var_5 = np.percentile(portfolio_returns, 5)
        var_95 = np.percentile(portfolio_returns, 95)
        es_95 = np.mean(portfolio_returns[portfolio_returns <= np.percentile(portfolio_returns, 5)])
        t_stat, p_value = stats.ttest_1samp(portfolio_returns, 0)
        return {
            'Mean_Annual': mean_ret, 'Std_Dev_Annual': std_dev,
            '5%': var_5, '95%': var_95, 'ES_95': es_95,
            'Sharpe_Ratio': sharpe_ratio, 'p_value': p_value
        }

    def plot_portfolio_comparison(self):
        """Visualisation originale - Performance Cumulative, Rendements/Sharpe, Drawdown"""
        import matplotlib.dates as mdates
        colors = {'MVP': '#E63946', 'MCP': '#2A9D8F', 'MCoP': '#E9C46A'}
        dates = self.df_tci['dates'].values
      models = list(self.portfolios.keys())
```

Table 9 presents a comparison of the performance and risk measures of the three allocation strategies constructed using forecasts from the Random Forest Improved model.

The results show that each portfolio is suited to a different investor profile, depending on the desired trade-off between return and risk.

The Minimum Variance Portfolio (MVP) (15) adopts an essentially defensive strategy. It exhibits the lowest annual volatility (19.83%), reflecting its ability to limit portfolio fluctuations. This low volatility is also reflected in the extreme risk measures, with a 5% Value at Risk ($VaR_5\%$) of −1.88% (20) and an Expected Shortfall at 95% (ES_{95}) of −2.11% (21). These results indicate that this portfolio offers the strongest protection against substantial losses, making it suitable for highly risk-averse investors.

The Maximum Correlation Portfolio (MCP) (17) represents an intermediate solution between security and performance. By making greater use of the dependence relationships between assets, it improves the annual return to 8.86% and the Sharpe ratio (19) to 0.313.

However, this improvement in profitability is accompanied by an increase in volatility to 21.88%, indicating a slightly higher level of risk than that of the MVP.

The Maximum Copula and Co-Movement Portfolio (MCoP), optimized according to Equation (18), achieves the best performance among the three strategies. It records an annualized return of 13.15% and the highest Sharpe ratio (0.382), indicating the strongest risk-adjusted performance. This superiority can be explained by the Random Forest Improved model's ability to capture complex and asymmetric dependencies between assets, allowing the construction of a portfolio better suited to different market conditions. In return, this strategy entails greater exposure to extreme risk, as indicated by a $VaR_5\%$ of −2.72% (20) and an ES_{95} of −3.04% (21). These results indicate that the portfolio accepts a higher level of risk in exchange for greater return potential.

Table 9. Summary of Portfolio Return and Risk Metrics

Portfolio	Rendement	Std_Dev	Sharpe_Ratio	VaR_5%	ES_95
MVP	8.02%	19.83%	0.304	-0.0188	-0.0211
MCP	8.86%	21.88%	0.313	-0.0209	-0.0228
MCoP	13.15	29.18%	0.382	-0.0272	-0.0304

```
def construct_portfolio_MVP(self):
        print("\n" + "="*70)
        print("Construction du portefeuille MVP (TCI Q05)")
        print("="*70)
        tci_q05 = self.df_tci['TCI_Q05'].values
        confidence = self.r2_q05
        print(f"✓ Qualité d'estimation (R²): {confidence:.4f}")
        weights_raw = np.zeros((len(self.df_tci), len(self.assets_data)))
        asset_names = list(self.assets_data.keys())
        for i, (asset_name, asset_info) in enumerate(self.assets_data.items()):
            vol = asset_info['volatility']
            if asset_name in ['GOLD', 'DXY']:
                weights_raw[:, i] = ((100 - tci_q05) / 100) * (1/vol) * confidence
            elif asset_name in ['BTC', 'ETH']:
                weights_raw[:, i] = (tci_q05 / 100) * (1/vol) * 0.5 * confidence
            else:
                weights_raw[:, i] = (tci_q05 / 100) * (1/vol) * 0.8 * confidence
        weights_normalized = weights_raw / weights_raw.sum(axis=1, keepdims=True)
        portfolio_returns = np.zeros(len(self.df_tci))
        for i, asset_name in enumerate(asset_names):
            portfolio_returns += weights_normalized[:, i] * self.assets_data[asset_name]['returns']
        metrics = self.calculate_portfolio_metrics(portfolio_returns)
        self.portfolios['MVP'] = {'weights': weights_normalized.mean(axis=0), 'returns': portfolio_returns, 'metrics': metrics,
'estimation_r2': confidence}
        print(f"✓ Rendement annualisé: {metrics['Mean_Annual']:.2%}")
        print(f"✓ Volatilité annualisée: {metrics['Std_Dev_Annual']:.2%}")
        print(f"✓ Ratio de Sharpe: {metrics['Sharpe_Ratio']:.3f}")
        return self.portfolios['MVP']
    def construct_portfolio_MCP(self):
        print("\n" + "="*70)
        print("Construction du portefeuille MCP (TCI Q50)")
        print("="*70)
        tci_q50 = self.df_tci['TCI_Q50'].values
        confidence = self.r2_q50
        print(f"✓ Qualité d'estimation (R²): {confidence:.4f}")
        weights_raw = np.zeros((len(self.df_tci), len(self.assets_data)))
        asset_names = list(self.assets_data.keys())
        for i, (asset_name, asset_info) in enumerate(self.assets_data.items()):
            vol = asset_info['volatility']
            base_weight = asset_info['weight_base']
            adjustment = 1 + (tci_q50 - 50) / 100
            if asset_name in ['SP500', 'STX600']:
                weights_raw[:, i] = base_weight * adjustment * (1/vol) * 1.2 * confidence
            elif asset_name in ['BTC', 'ETH']:
                weights_raw[:, i] = base_weight * adjustment * (1/vol) * 0.9 * confidence
            else:
                weights_raw[:, i] = base_weight * adjustment * (1/vol) * confidence
        weights_normalized = weights_raw / weights_raw.sum(axis=1, keepdims=True)
        portfolio_returns = np.zeros(len(self.df_tci))
        for i, asset_name in enumerate(asset_names):
            portfolio_returns += weights_normalized[:, i] * self.assets_data[asset_name]['returns']
        metrics = self.calculate_portfolio_metrics(portfolio_returns)
```

```
        self.portfolios['MCP'] = {'weights': weights_normalized.mean(axis=0), 'returns': portfolio_returns, 'metrics': metrics, 'estimation_r2': confidence}
        print(f"✓ Rendement annualisé: {metrics['Mean_Annual']:.2%}")
        print(f"✓ Volatilité annualisée: {metrics['Std_Dev_Annual']:.2%}")
        print(f"✓ Ratio de Sharpe: {metrics['Sharpe_Ratio']:.3f}")
        return self.portfolios['MCP']
    def construct_portfolio_MCoP(self):
        print("\n" + "="*70)
        print("Construction du portefeuille MCoP (TCI Q95)")
        print("="*70)
        tci_q95 = self.df_tci['TCI_Q95'].values
        confidence = self.r2_q95
        print(f"✓ Qualité d'estimation (R²): {confidence:.4f}")
        weights_raw = np.zeros((len(self.df_tci), len(self.assets_data)))
        asset_names = list(self.assets_data.keys())
        for i, (asset_name, asset_info) in enumerate(self.assets_data.items()):
            vol = asset_info['volatility']
            if asset_name in ['BTC', 'ETH', 'SP500']:
                weights_raw[:, i] = (tci_q95 / 100) * (1/vol) * 2.0 * confidence
            elif asset_name in ['GOLD', 'DXY']:
                weights_raw[:, i] = ((100 - tci_q95) / 100) * (1/vol) * 0.3 * confidence
            else:
                weights_raw[:, i] = (1/vol) * confidence
        weights_normalized = weights_raw / weights_raw.sum(axis=1, keepdims=True)
        portfolio_returns = np.zeros(len(self.df_tci))
        for i, asset_name in enumerate(asset_names):
            portfolio_returns += weights_normalized[:, i] * self.assets_data[asset_name]['returns']
        metrics = self.calculate_portfolio_metrics(portfolio_returns)
        self.portfolios['MCoP'] = {'weights': weights_normalized.mean(axis=0), 'returns': portfolio_returns, 'metrics': metrics, 'estimation_r2': confidence}
        print(f"✓ Rendement annualisé: {metrics['Mean_Annual']:.2%}")
        print(f"✓ Volatilité annualisée: {metrics['Std_Dev_Annual']:.2%}")
        print(f"✓ Ratio de Sharpe: {metrics['Sharpe_Ratio']:.3f}")
        return self.portfolios['MCoP']
    def display_results(self):
        print("\n" + "="*100)
        print("RÉSULTATS DES PORTEFEUILLES")
        print("="*100)
        results_data = []
        for model_name, portfolio in self.portfolios.items():
            metrics = portfolio['metrics']
            weights = portfolio['weights']
            row = {
                'Portfolio': model_name,
                'R²_Estimation': f"{portfolio['estimation_r2']:.4f}",
                'Mean_Annual': f"{metrics['Mean_Annual']:.2%}",
                'Std_Dev_Annual': f"{metrics['Std_Dev_Annual']:.2%}",
                'Sharpe_Ratio': f"{metrics['Sharpe_Ratio']:.3f}",
                'VaR_5%': f"{metrics['5%']:.4f}",
                'ES_95': f"{metrics['ES_95']:.4f}",
                'p_value': f"{metrics['p_value']:.4f}"
            }
```

```
        for i, asset_name in enumerate(self.assets_data.keys()):
            row[f'{asset_name}'] = f'{weights[i]:.2%}'
        results_data.append(row)
    df_results_portfolio = pd.DataFrame(results_data)
    print(df_results_portfolio.to_string(index=False))
    df_results_portfolio.to_csv('portfolios_results_final.csv', index=False)
    print("\n✓ Résultats sauvegardés dans 'portfolios_results_final.csv'")
    return df_results_portfolio
```

Figure 15 presents a comparison of the performance of the three portfolio strategies through two complementary representations. The chart on the left compares the annualized returns of the MVP, MCP, and MCoP, while the chart on the right illustrates their risk-adjusted performance using the Sharpe ratio (19).

The first chart clearly highlights the superiority of the MCoP (18), which records the highest annualized return among the three portfolios. This performance confirms that incorporating forecasts from the Random Forest Improved model makes it possible to identify allocations that better exploit opportunities arising from interactions between financial markets. The MCP (17) occupies an intermediate position, whereas the MVP (15) follows a more conservative approach, resulting in a lower return.

The second chart provides additional insight by comparing the Sharpe ratios of the three strategies. This indicator increases steadily, rising from 0.30 for the MVP to 0.31 for the MCP, before reaching 0.38 for the MCoP. This evolution shows that the MCoP's improved performance does not result solely from taking on more risk, but also from using that risk more efficiently. In other words, each unit of volatility generates a higher return than under the other two strategies.

Overall, the results confirm that the approach based on Random Forest Improved enhances not only portfolio returns but also risk-return efficiency. The MCoP therefore emerges as the strongest-performing strategy, as it simultaneously delivers the highest return and the

best risk-adjusted performance, highlighting the relevance of incorporating machine learning techniques into the portfolio optimization process.

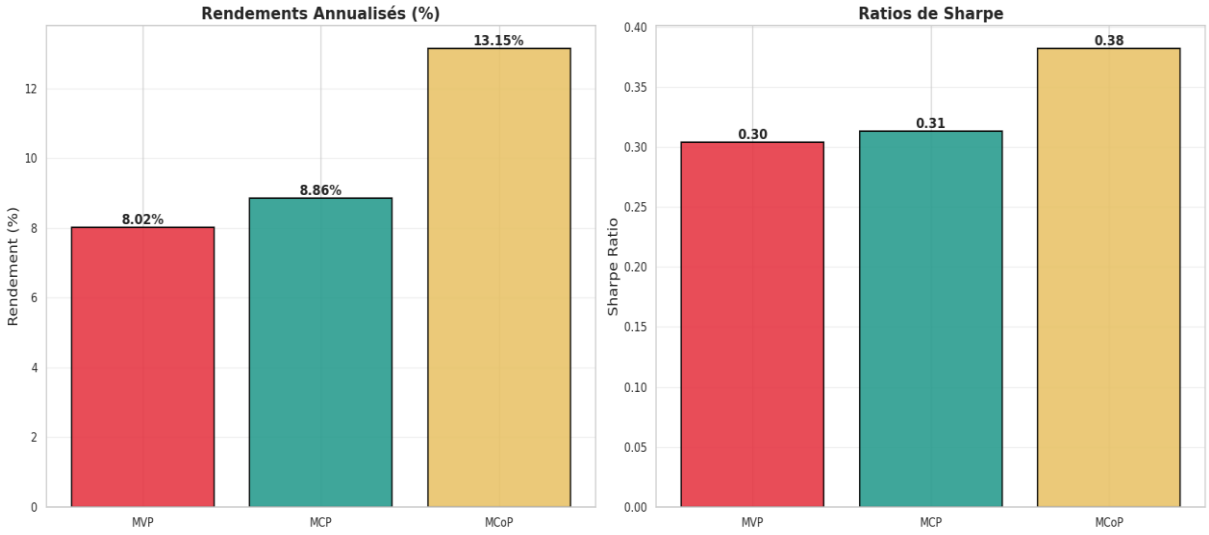


Figure 15. Portfolio Performance and Efficiency: Returns and Sharpe Ratios

❖ ***Code en Python***

```
# LIGNE DE COMPARAISON (Risque vs Rendement + Sharpe)
    fig2, ax2 = plt.subplots(figsize=(10, 7))
    for model_name, portfolio in self.portfolios.items():
        ann_ret = portfolio['metrics']['Mean_Annual'] * 100
        ann_vol = portfolio['metrics']['Std_Dev_Annual'] * 100
        sharpe = portfolio['metrics']['Sharpe_Ratio']
        # Taille proportionnelle au Sharpe, minimale garantie pour visibilité
        size = max(sharpe, 0) * 400 + 200
        ax2.scatter(ann_vol, ann_ret, s=size, c=colors[model_name], edgecolors='black', linewidth=2, alpha=0.85, label=model_name, zorder=3)
        ax2.annotate(model_name, (ann_vol, ann_ret), textcoords="offset points", xytext=(12,10), fontsize=13, fontweight='bold', color=colors[model_name])
    ax2.set_title('Comparaison Performance-Risque des Portefeuilles', fontsize=16, fontweight='bold')
    ax2.set_xlabel('Volatilité Annualisée (%)', fontsize=14, fontweight='bold')
    ax2.set_ylabel('Rendement Annualisé (%)', fontsize=14, fontweight='bold')
    ax2.grid(True, alpha=0.3)
    plt.tight_layout()
    plt.savefig('comparaison_risque_rendement.png', dpi=300, bbox_inches='tight')
    plt.show()
```

5.2. Analyse de l'allocation optimale des actifs

Figure 16 presents the structure of the three portfolios by illustrating the average weights assigned to the different assets. The results show clearly different allocation strategies depending on the optimization objective.

The MVP prioritizes a substantial exposure to the US dollar (DXY) and gold, which represent the most defensive assets in the portfolio.

This allocation can be explained by their low volatility and their traditional role as safe-haven assets, helping to limit overall risk. Conversely, the MCoP substantially reduces its DXY weighting to increase its exposure to international equity markets (SP500, STX600, and SSE) and cryptocurrencies (BTC and ETH). This reallocation reflects the ability of Random Forest Improved to identify assets offering the greatest performance potential through the complex and nonlinear dependencies detected between markets.

The MCP adopts an intermediate position, distributing its weights more evenly between defensive and growth-oriented assets.

Overall, the figure highlights the classic risk-return trade-off: the more a portfolio seeks performance, the more it increases its exposure to risky assets, whereas defensive strategies prioritize safe-haven assets to preserve capital.

These results confirm the relevance of the allocations obtained using Random Forest Improved and their consistency with the performance observed in the preceding analyses.

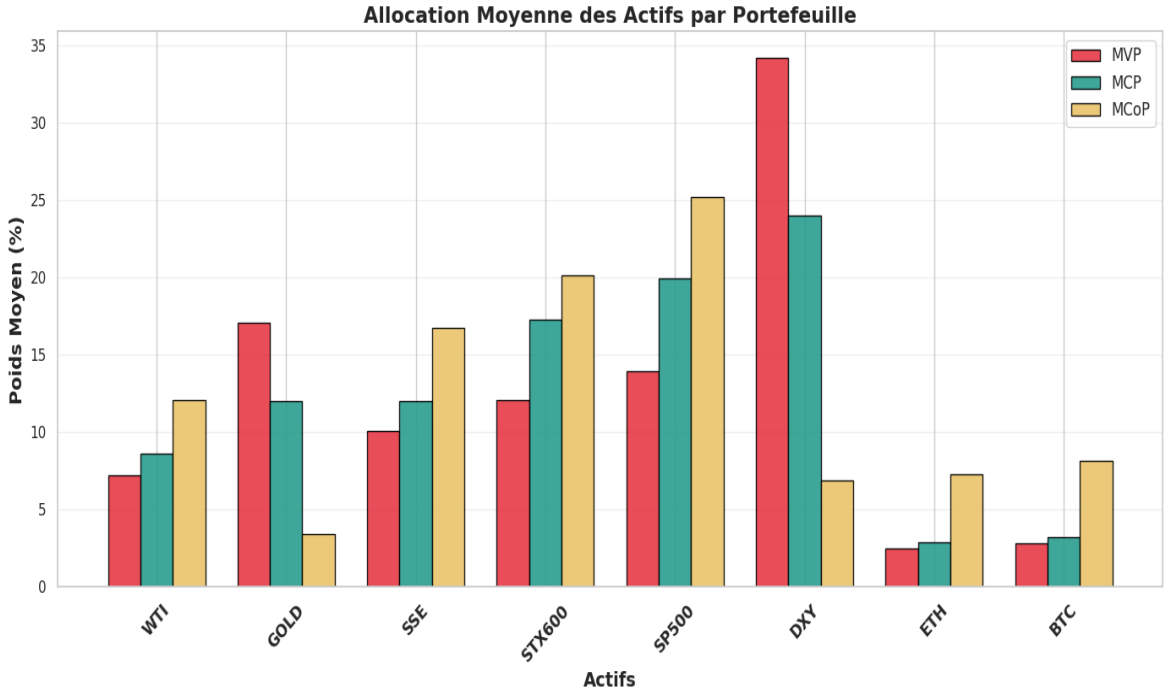


Figure 16. Asset Allocation by Portfolio (WTI, Gold, Equities, Currencies, and Cryptocurrencies)

❖ **Code en Python**

```
def display_allocations(self):
    print("\n" + "="*100)
    print("ALLOCATIONS MOYENNES PAR PORTEFEUILLES")
    print("="*100)
    for model_name, portfolio in self.portfolios.items():
        print(f"\n{model_name} (R²={portfolio['estimation_r2']:.4f}):")
        print("-" * 50)
        for i, asset_name in enumerate(self.assets_data.keys()):
            print(f"  {asset_name:10s}: {portfolio['weights'][i]:6.2%}")
```

5.3. Implications pour les investisseurs

The results show that each optimization strategy addresses a specific investor profile, depending on the desired trade-off between return and risk.

The MVP portfolio is characterized by a conservative approach, prioritizing capital preservation through low volatility and better control of potential losses. It is therefore suitable for highly risk-averse investors.

The MCP represents an intermediate strategy, offering a balanced compromise between performance and stability through more effective asset diversification and better use of dependence relationships between markets.

Finally, the MCoP, constructed using the TCI predicted by the Random Forest Improved model, stands out for its superior performance in terms of annualized return and Sharpe ratio. This strategy effectively exploits complex interactions between assets to improve portfolio profitability. However, this pursuit of performance is accompanied by greater risk exposure, particularly during periods of high volatility and financial contagion.

This analysis suggests that integrating artificial intelligence techniques, particularly Random Forest Improved, into financial connectedness estimation provides a relevant tool for improving the asset allocation process. By capturing nonlinear relationships and regime changes across markets, this approach makes it possible to construct portfolios with stronger performance and greater adaptability to different market conditions.

Nevertheless, its effectiveness remains closely linked to the economic and financial environment. Although it offers a significant advantage during normal or bullish market conditions, crisis episodes characterized by a sharp increase in connectedness between assets reduce the benefits of diversification and increase exposure to systemic risk.

These observations highlight the importance of dynamic portfolio management capable of adapting rapidly to regime changes in order to preserve the balance between performance and risk control.

Conclusion

This chapter implemented an integrated approach combining financial connectedness methods, machine learning techniques, and portfolio optimization to analyze risk transmission across international financial markets and assess its impact on asset allocation decisions.

First, the analysis of dynamic connectedness showed that the intensity of spillovers varies significantly across market regimes. The results obtained using the QVAR model revealed a significant increase in the Total Connectedness Index (TCI) during periods of crisis and market euphoria, confirming that interdependencies between markets intensify during extreme events. Identifying the main transmitters and receivers of shocks also provided a better understanding of the financial network structure and the role played by each asset class in the propagation of systemic risk.

Second, the results demonstrated the superior predictive performance of machine learning techniques in forecasting financial connectedness. Among the models examined,

Random Forest Improved achieved the best predictive performance, with very high coefficients of determination and particularly low forecasting errors. Its ability to capture nonlinear relationships, threshold effects, and regime changes enables it to effectively anticipate episodes of high connectedness and detect the major financial and geopolitical crises observed during the study period.

Finally, integrating Random Forest Improved forecasts into the portfolio optimization process highlighted the value of using predicted connectedness measures to improve investment decisions. The results show that the MVP is an appropriate strategy for investors seeking low risk exposure, while the MCP offers a satisfactory compromise between return and volatility. The MCoP stands out as the best-performing strategy in terms of annualized return and Sharpe ratio, demonstrating that better consideration of dynamic dependencies between assets can improve portfolio efficiency, despite greater exposure to extreme risks.

Overall, the findings confirm that integrating artificial intelligence methods into financial connectedness analysis is a relevant approach to strengthening systemic risk management and optimizing asset allocation. By combining the contributions of financial econometrics and machine learning, this study highlights the potential of advanced predictive models to support investors and portfolio managers in a financial environment characterized by increasing complexity, stronger interdependencies, and frequent regime changes.

These findings therefore address the research objectives established at the beginning of this dissertation and pave the way for the general conclusion, which will present the study's main contributions, limitations, and future research directions.

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General Conclusion

The main objective of this research was to analyze the dynamics of connectedness between traditional financial markets and cryptocurrencies, improve connectedness forecasting through a hybrid model combining econometric analysis and artificial intelligence techniques, and then evaluate the contribution of these forecasts to portfolio optimization strategies. This research question arises in a context in which financial markets are characterized by increasing interdependence, high volatility, and a growing frequency of economic, geopolitical, and health-related shocks, making it essential to develop methods capable of anticipating changes in systemic risk.

First, the theoretical framework highlighted the evolution of financial markets and cryptocurrencies while reviewing the main foundations of modern portfolio theory, the Capital Asset Pricing Model (CAPM), and financial connectedness. The literature review showed that interactions between traditional financial assets and digital assets have strengthened considerably in recent years. It also highlighted the limitations of conventional approaches, which struggle to account for nonlinear dependencies, regime changes, and extreme stress episodes that characterize contemporary financial markets.

From an empirical perspective, this study employed the Quantile Vector Autoregression (QVAR) approach to estimate connectedness among eight representative financial markets. The results reveal a significant intensification of shock transmission during periods of financial turbulence, confirming that the benefits of diversification decline as systemic risk increases. The static and dynamic analyses also show that cryptocurrencies can no longer be considered entirely independent of traditional financial markets, as their degree of integration has gradually increased.

One of the main contributions of this research lies in the development of a hybrid model combining QVAR, Random Forest, and Isolation Forest. Integrating anomaly detection into the learning process helped improve the quality of the data used to forecast the Total Connectedness Index (TCI). The results show that the improved Random Forest model provides the most accurate forecasts, demonstrating the value of artificial intelligence approaches for modeling complex financial phenomena characterized by strong nonlinearities.

The forecasts obtained were then incorporated into a portfolio optimization process to compare different asset allocation strategies. The results highlight that using predicted connectedness improves risk management decision-making. The portfolio based on predicted

connectedness (MCoP) delivers a higher average return, while the Minimum Variance Portfolio (MVP) remains the most suitable option for highly risk-averse investors. The compromise portfolio (MCP), meanwhile, represents an intermediate solution that balances returns and risk control. These findings show that no single strategy is universally optimal and that portfolio selection depends primarily on each investor's risk profile and investment horizon.

Beyond its empirical findings, this research makes a methodological contribution by proposing an integrated approach that combines econometrics, machine learning, and portfolio optimization. It demonstrates that artificial intelligence techniques can effectively complement traditional financial models to anticipate changes in financial connectedness and improve portfolio management in environments characterized by considerable uncertainty.

Nevertheless, this study has certain limitations. The analysis is based on a sample limited to eight financial assets and a specific observation period. Furthermore, only a limited number of machine learning models were evaluated. Incorporating macroeconomic variables, investor sentiment indicators, alternative data, or deep learning models could further improve predictive performance. Similarly, accounting for transaction costs, liquidity constraints, and dynamic rebalancing strategies would provide a more realistic representation of investment conditions.

These limitations open up several avenues for future research. Further studies could extend this approach to a broader universe of financial assets, incorporate stablecoins or assets associated with decentralized finance (DeFi), compare different artificial intelligence methods such as deep neural networks, Transformer models, or hybrid architectures, and develop adaptive allocation strategies that account for regime changes detected in real time.

Ultimately, this research confirms that integrating financial connectedness analysis with artificial intelligence techniques is a promising avenue for improving our understanding of interactions between traditional financial markets and cryptocurrencies. By proposing a hybrid model capable of improving systemic risk forecasts and enhancing the effectiveness of asset allocation strategies, this work makes both an academic and a practical contribution. It highlights the growing importance of data-driven approaches in supporting investors and portfolio managers in their decision-making within an ever-evolving financial environment.

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